DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015

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Register No : 240701137

Year / Branch : II Year / B.E CSE

Semester : IV

Academic Year : 2024-2028

INDEX

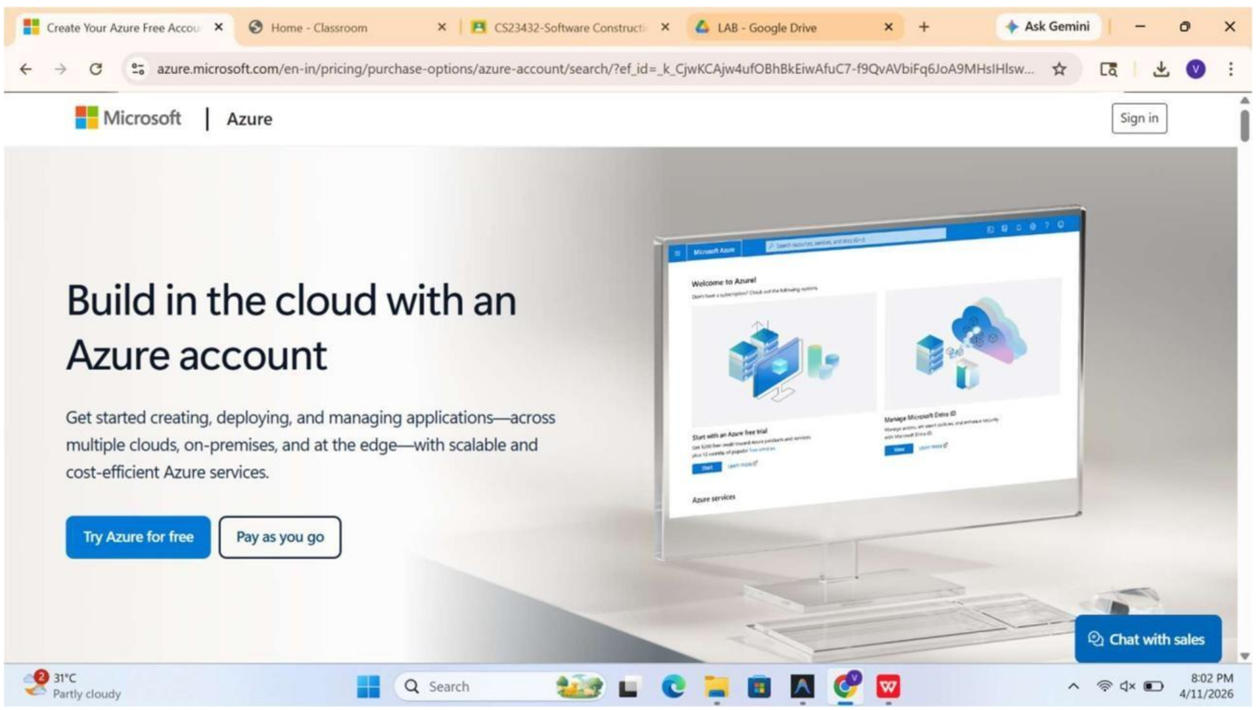
EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP

Aim:

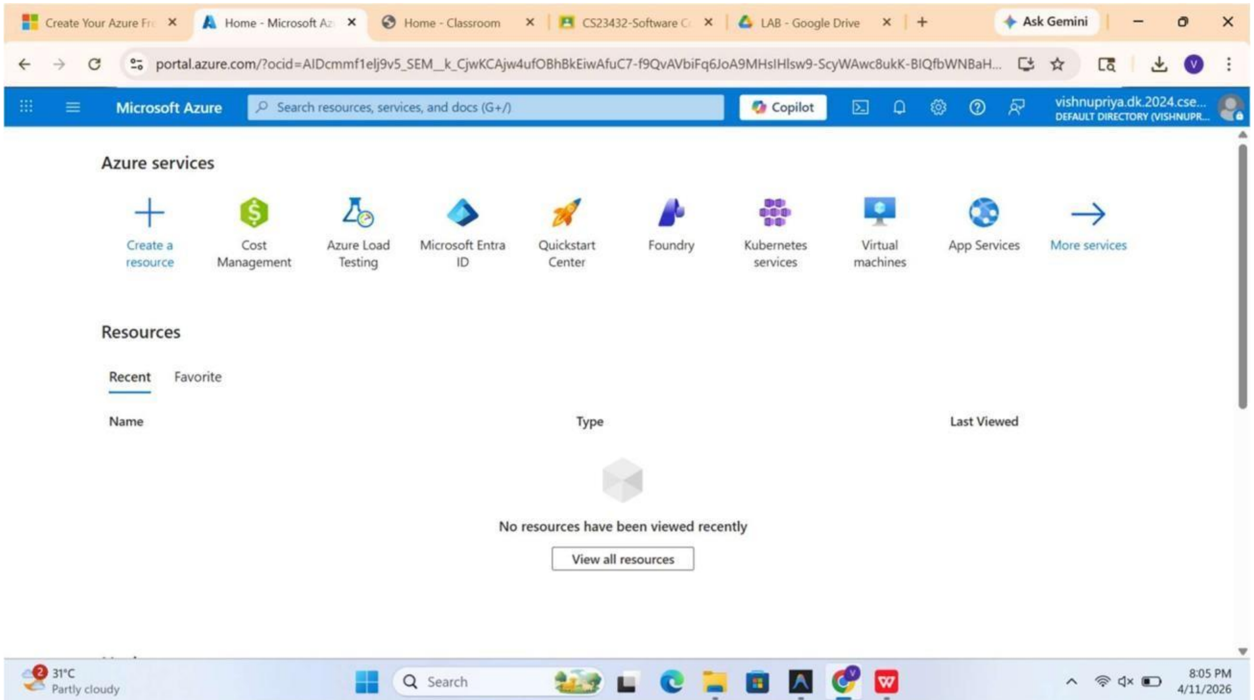
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

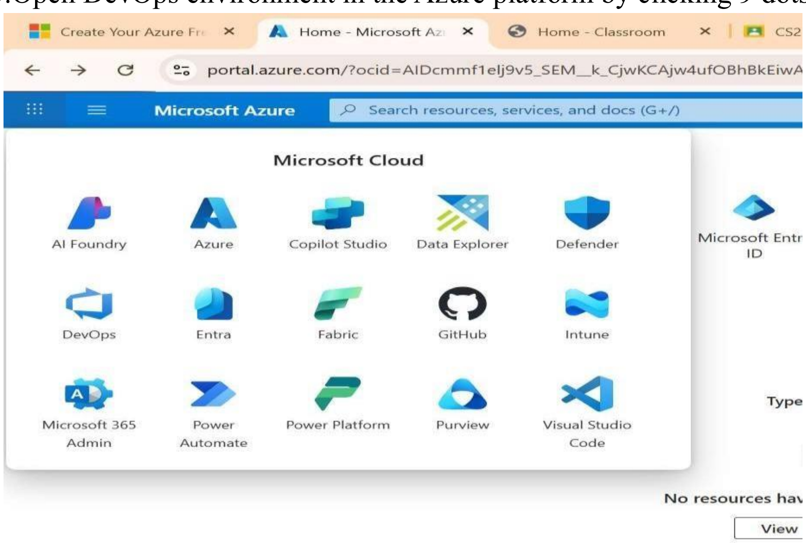
1.Open your web browser and go to the Azure website: https://azure.microsoft.com/en-us/getstarted/azure-portal.
Sign in using your Microsoft account credentials.
If you don't have a Microsoft account, you can create one here: https://signup.live.com/?lic=1



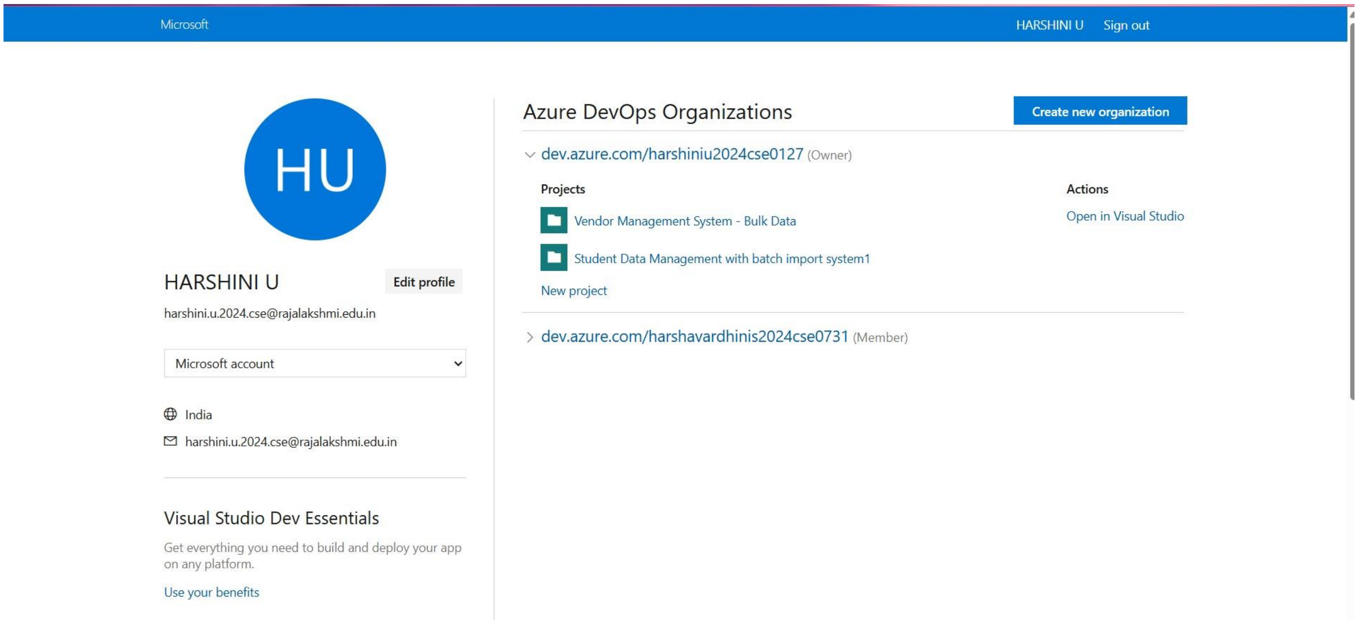
2.Azure home page

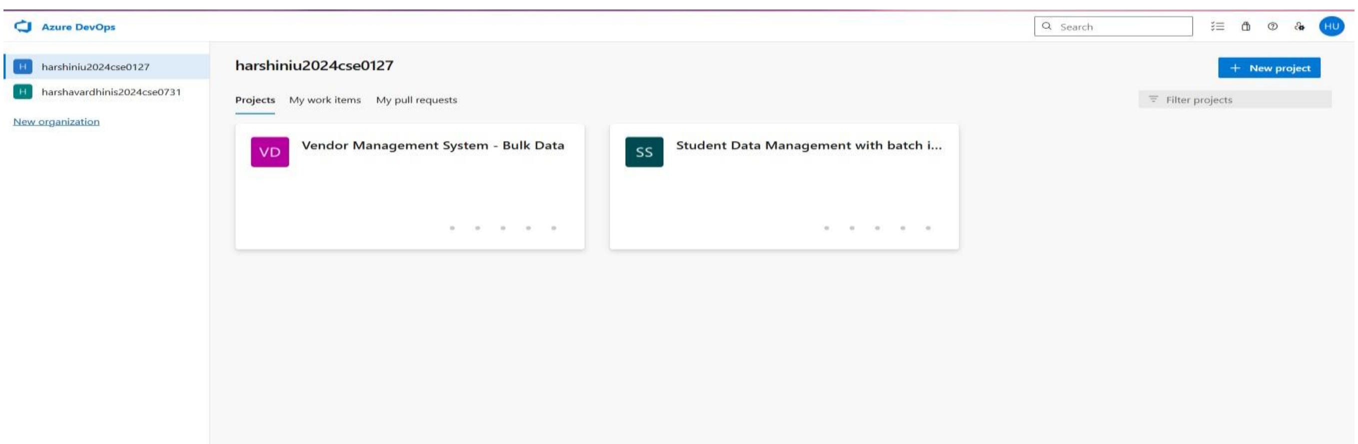


3.Open DevOps environment in the Azure platform by clicking 9 dots



4.Click on the DevOps icon and create an organization and you should be taken to the Azure DevOps Organization Home page.





Result:

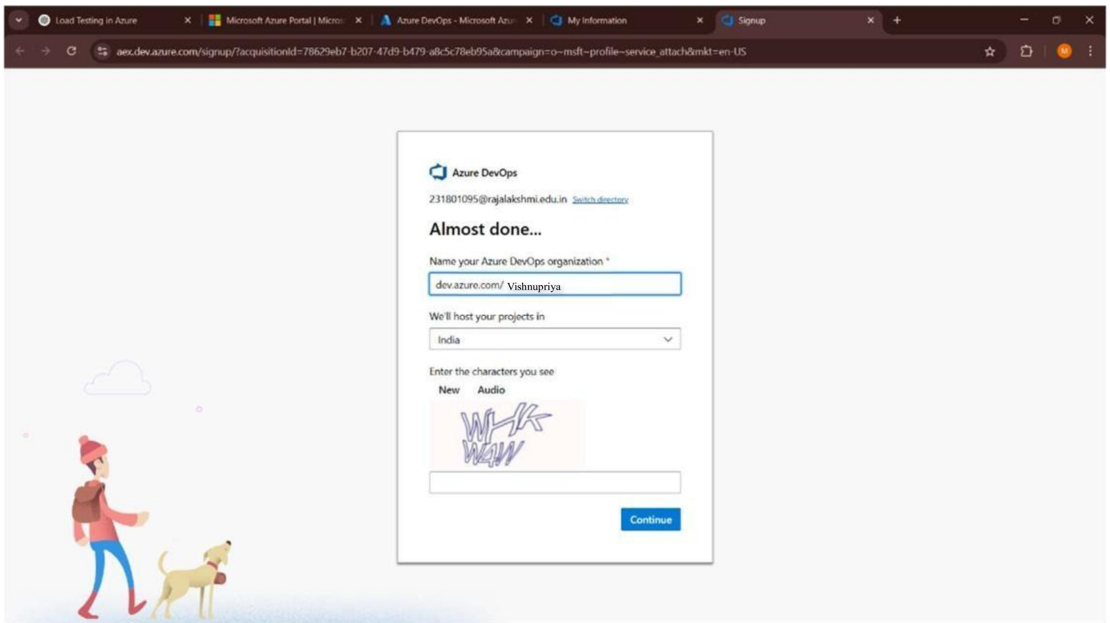
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY VENDOR MANAGEMENT SYSTEM WITH BULK DATA ENTRY

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

a. After the organization is set up, you’ll need to create your first project. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization’s Home page, click on the New Project button.

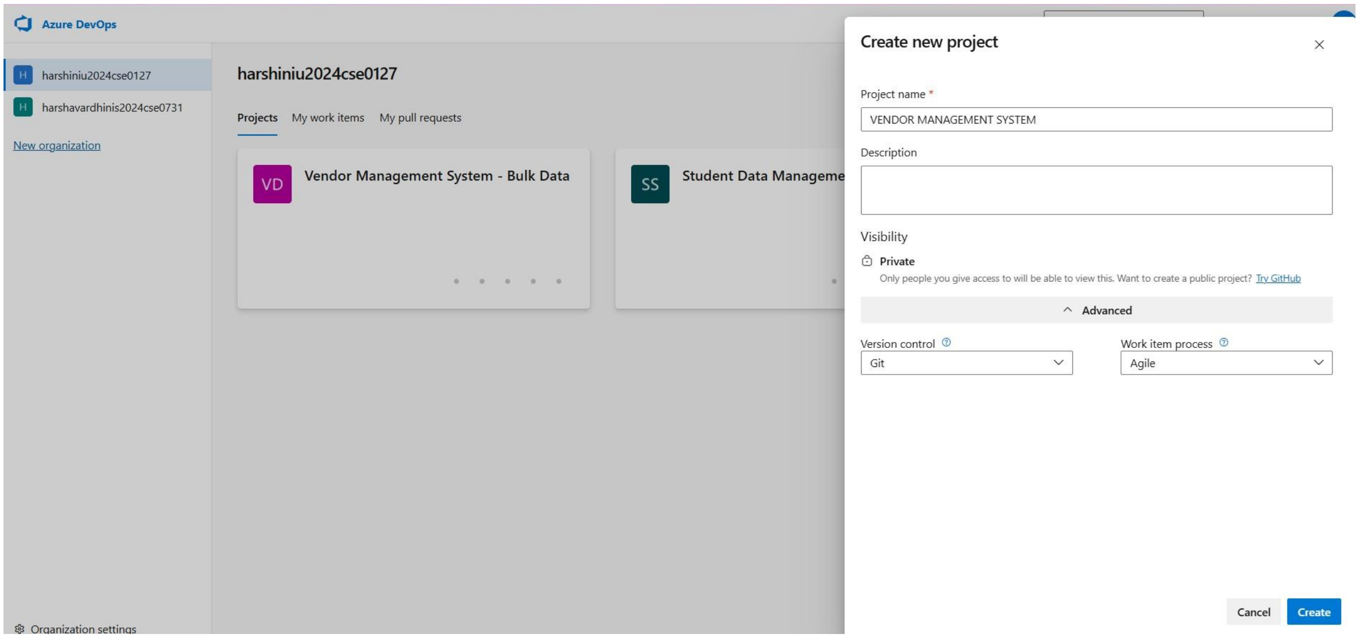
c. Enter the project name, description, and visibility options:

Name: Choose a name for the project (e.g., LMS).

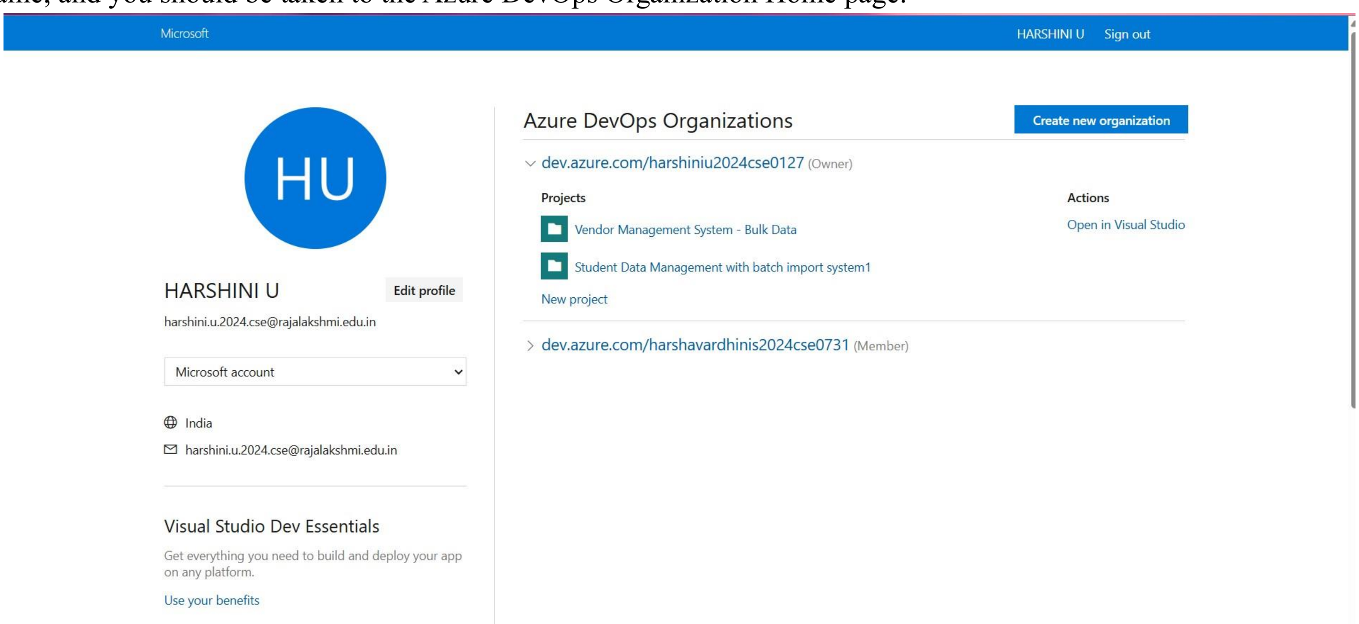
Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be Private (accessible only to those invited) or Public (accessible to anyone).

d. Once you’ve filled out the details, click Create to set up your first project.



3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



4.Project dashboard

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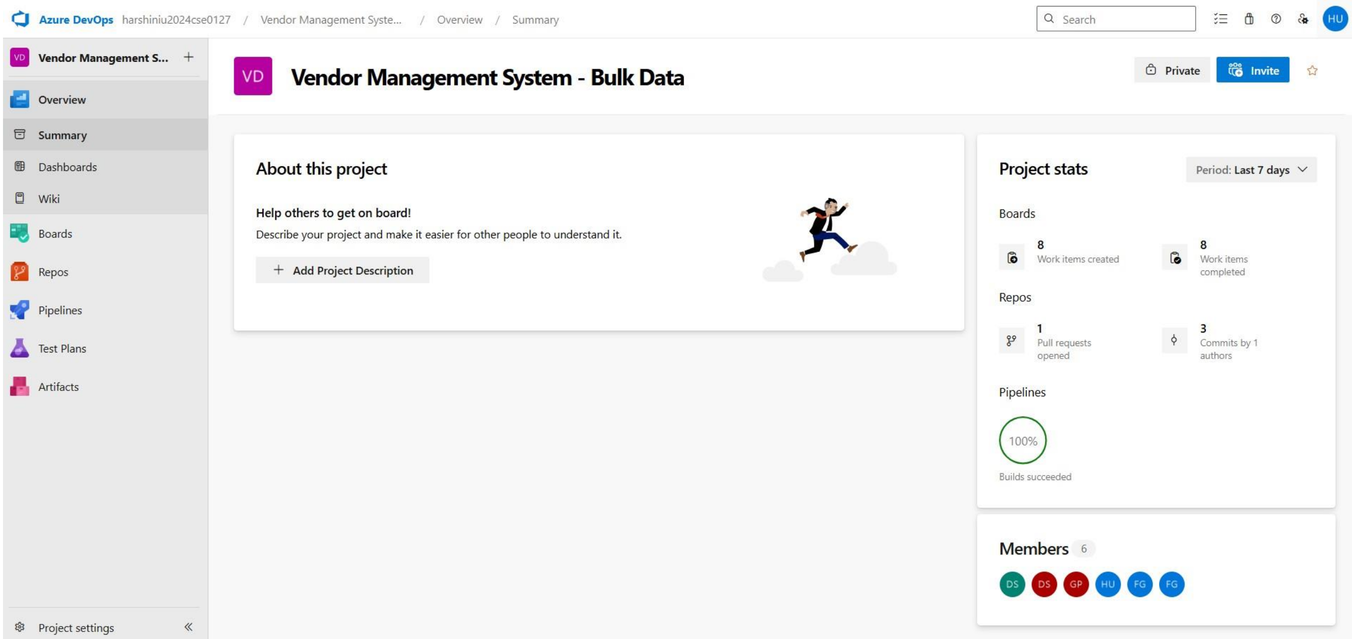
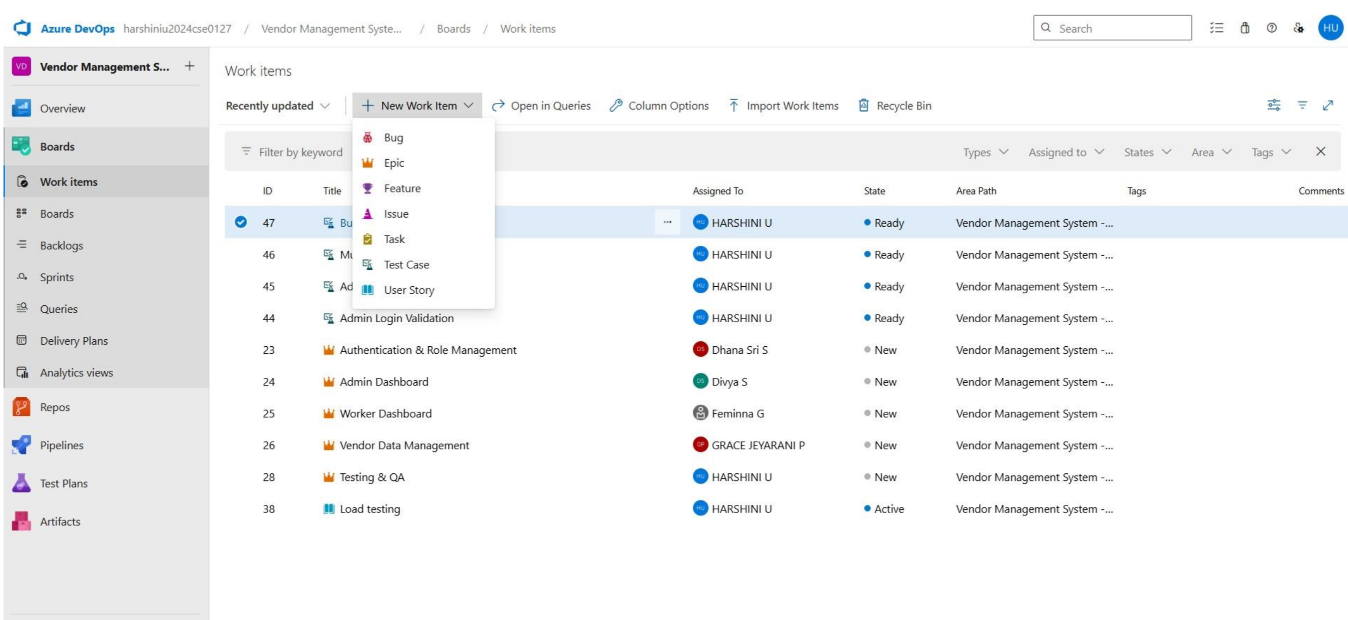


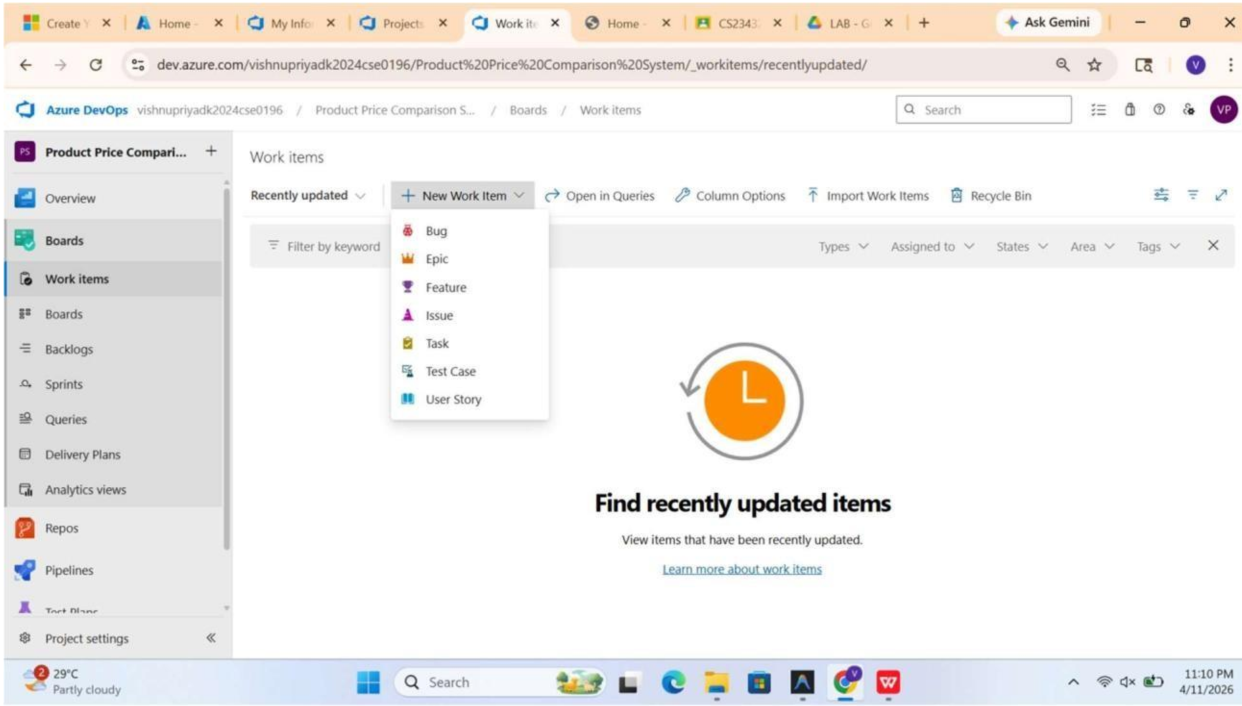
5.To manage user stories:

a. From the left-hand navigation menu, click on Boards. This will take you to the main Boards page, where you can manage work items, backlogs, and sprints.

b. On the work items page, you'll see the option to Add a work item at the top. Alternatively,

you can find a + button or Add New Work Item depending on the view you're in. From the Add a work item dropdown, select User Story. This will open a form to enter details for the new User Story.





Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

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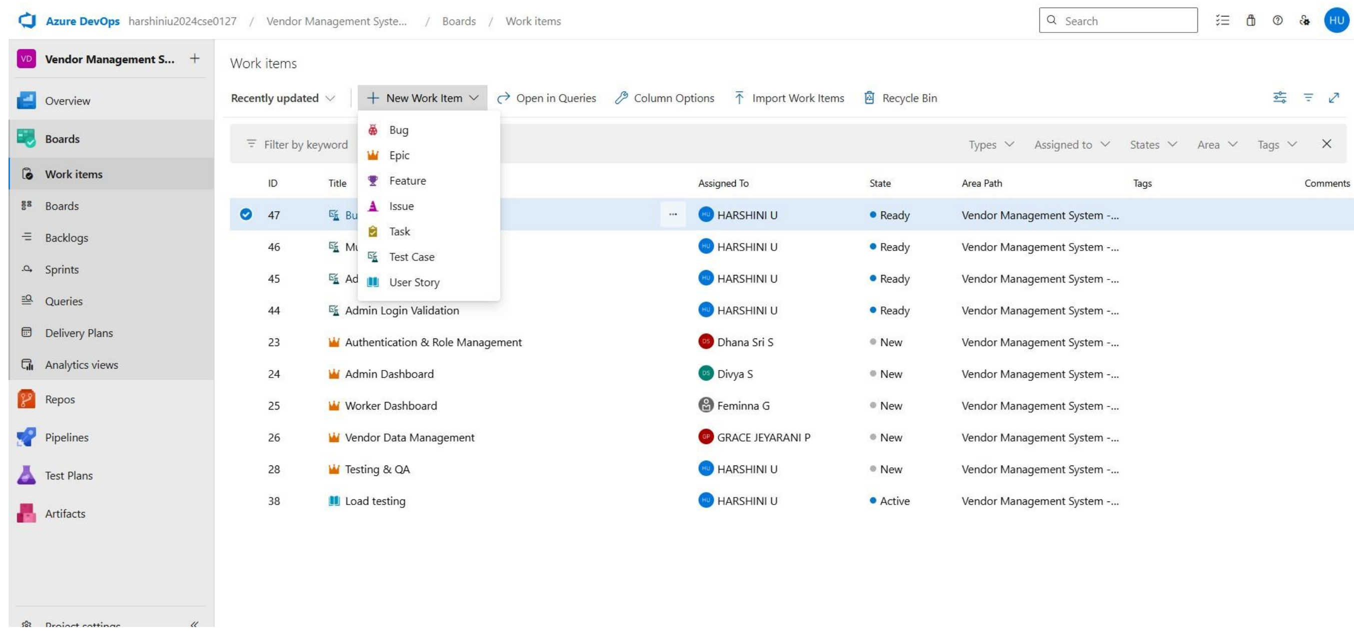
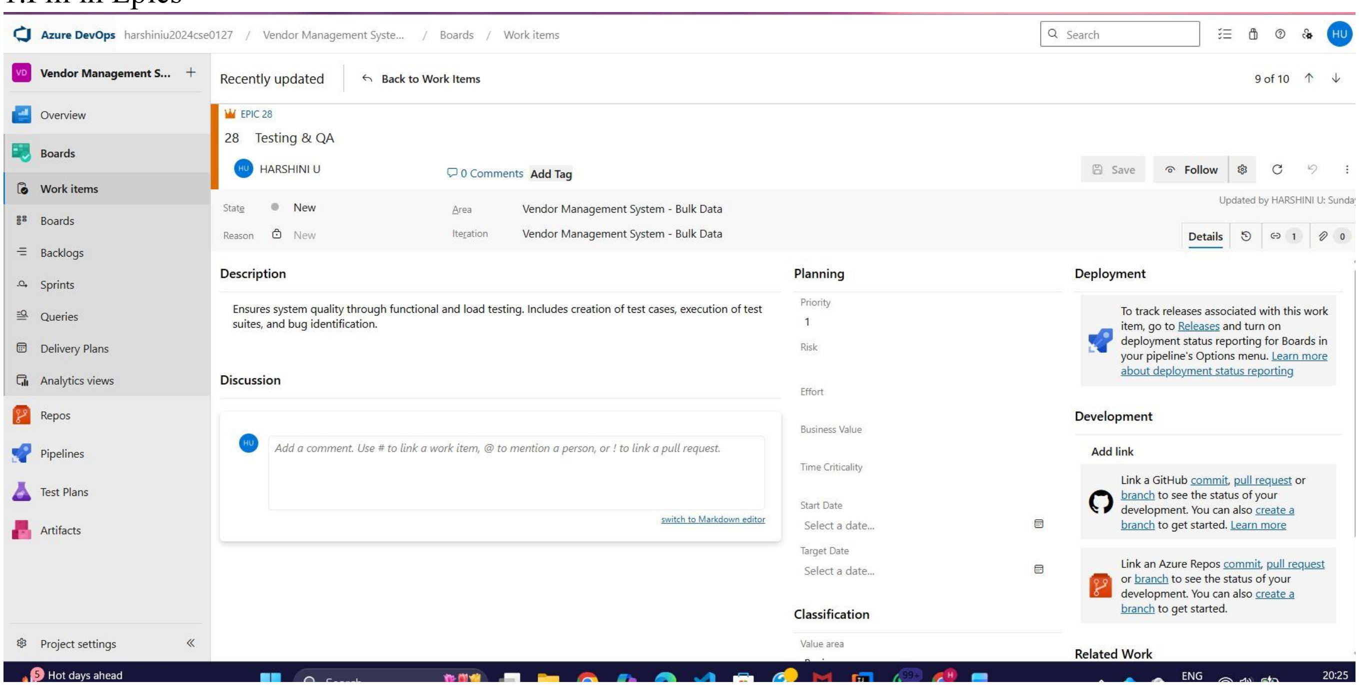
EXP NO : 3 F	SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

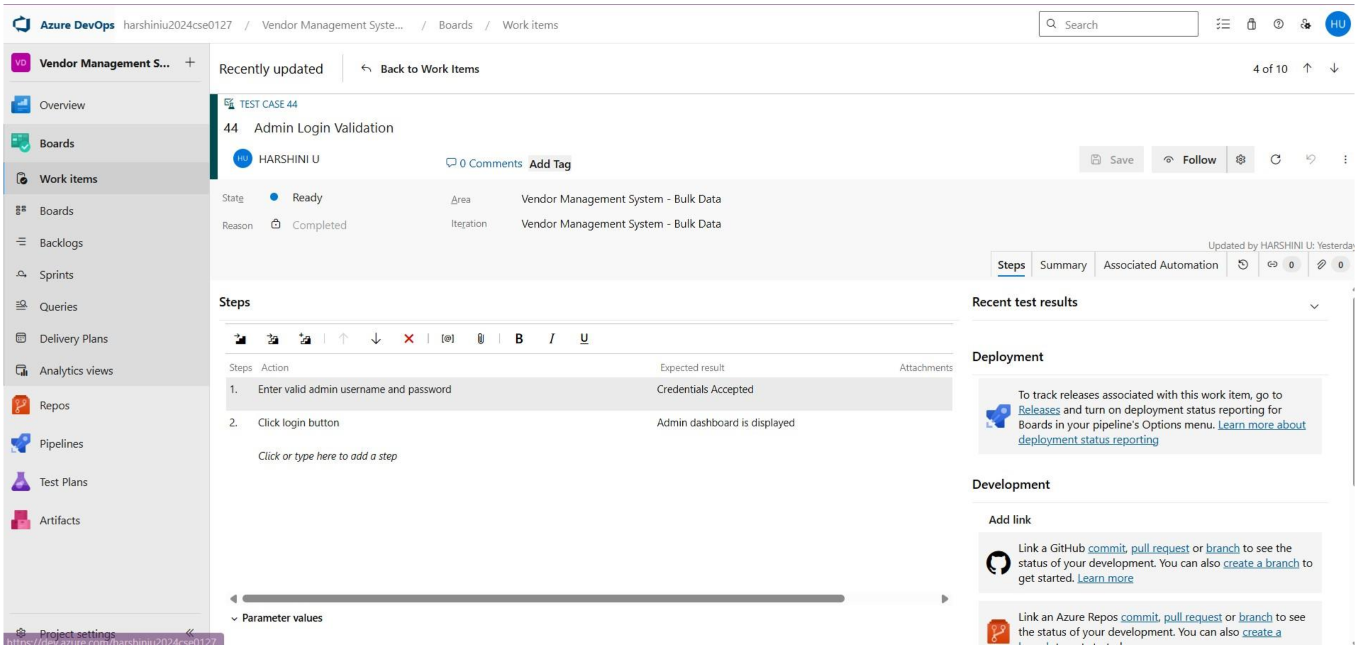
To learn about how to create epics, user story, features, backlogs for your assigned project.



1.Fill in Epics



3.Fill in User Story Details



Result:

Thus, the creation of epics, features, user story and task has been created successfully.

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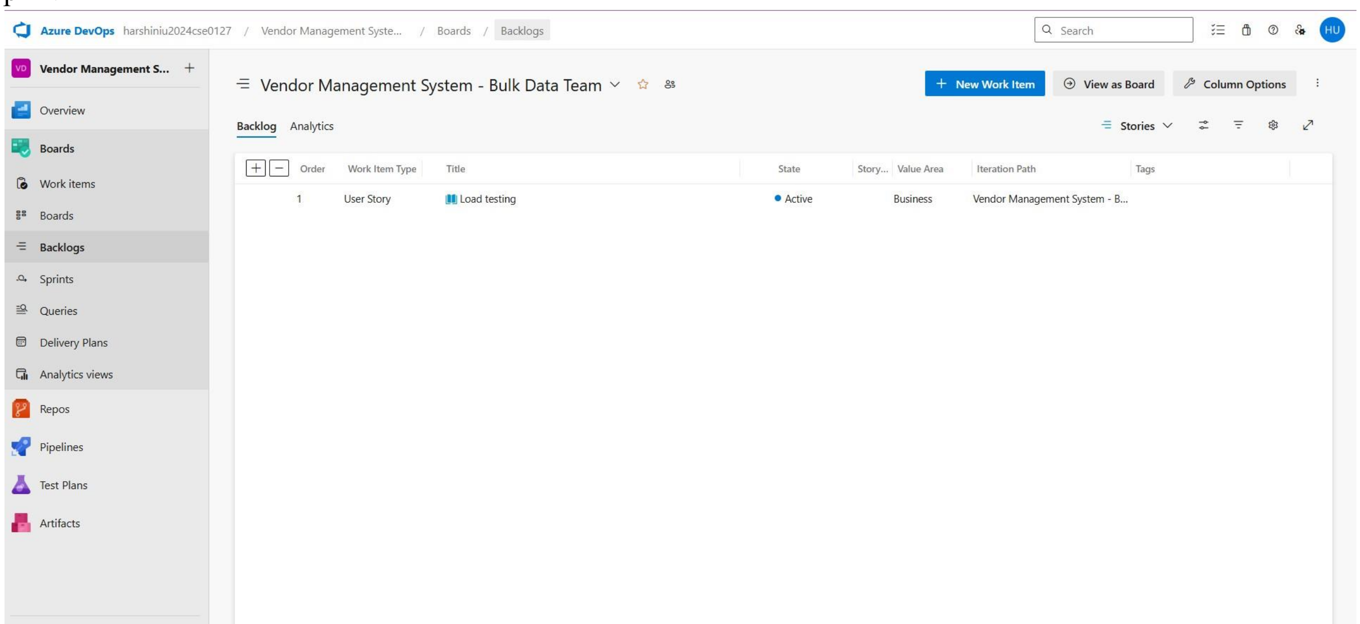
EXP NO: 4 F	SPRINT PLANNING

Aim:

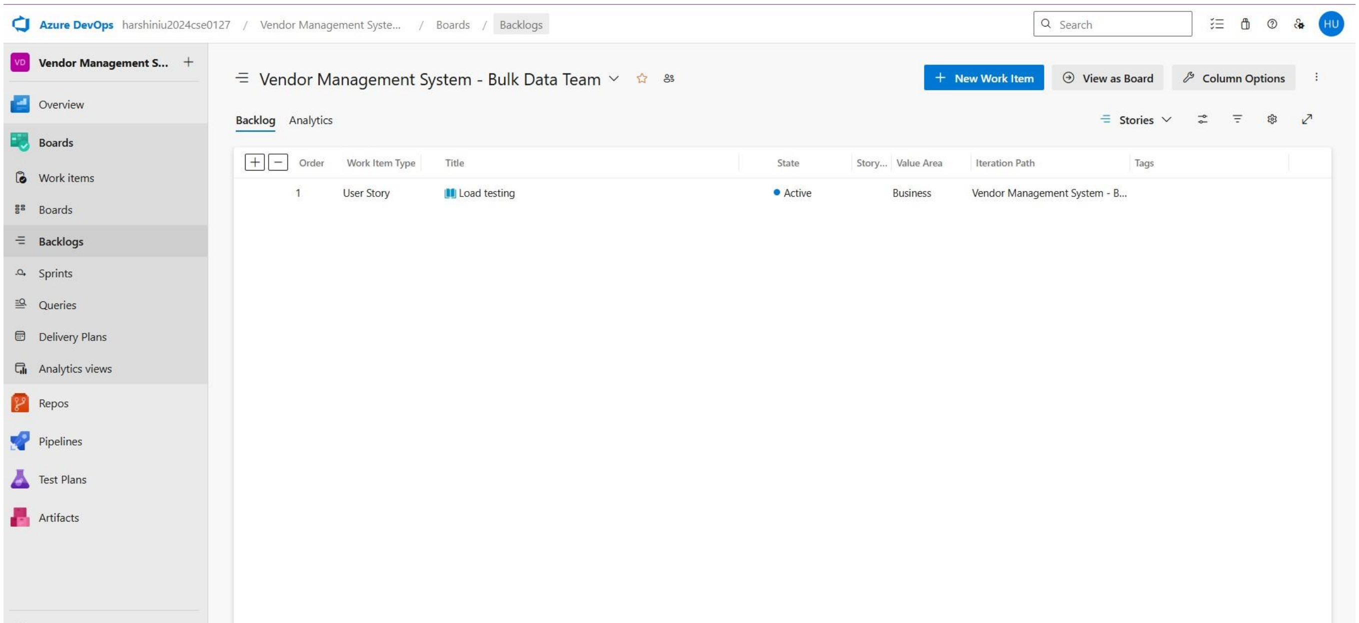
To assign user story to specific sprint for the VENDOR MANAGEMENT SYSTEM Project.

Sprint Planning

Sprint 1



Sprint 2



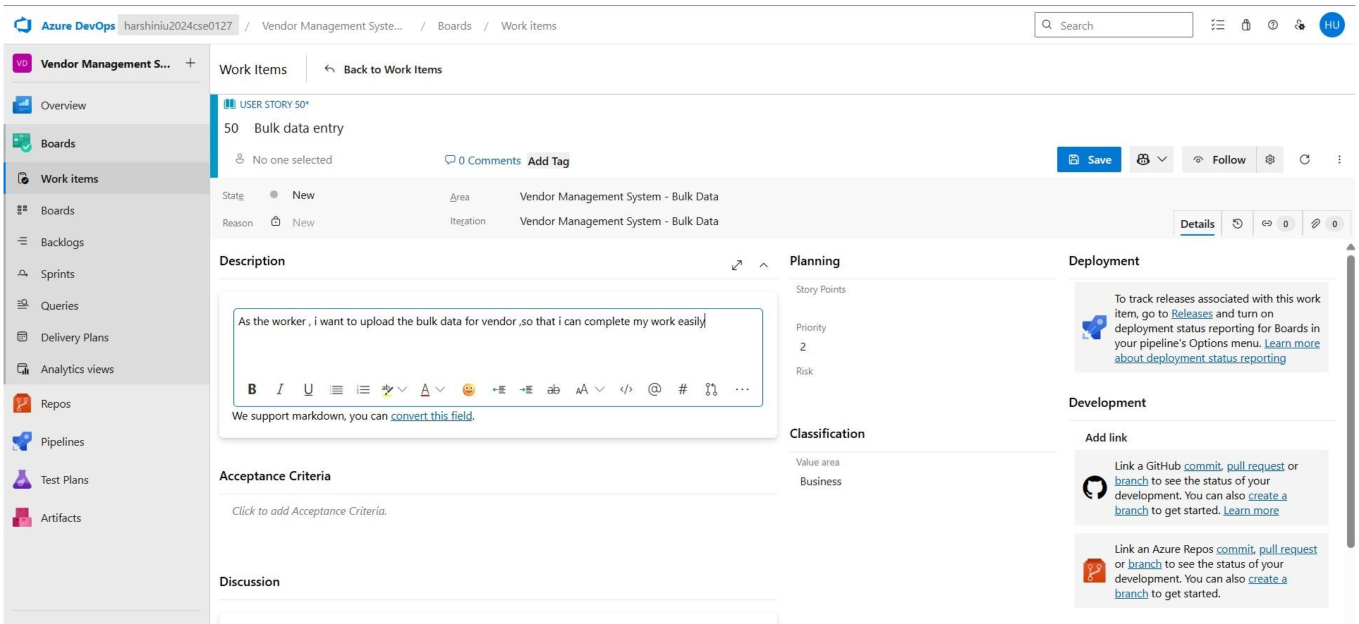
Result:

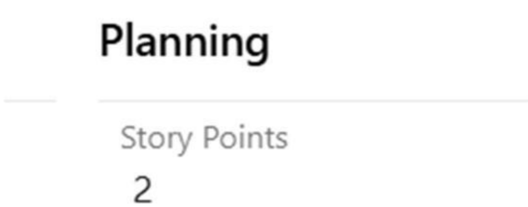
The Sprints are created for the VENDOR MANAGEMENT SYSTEM Project.

EXPNO:5	POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - the Product price Comparison System Project. Poker Estimation





Result:

The Estimation/Story Points is created for the project using Poker Estimation.

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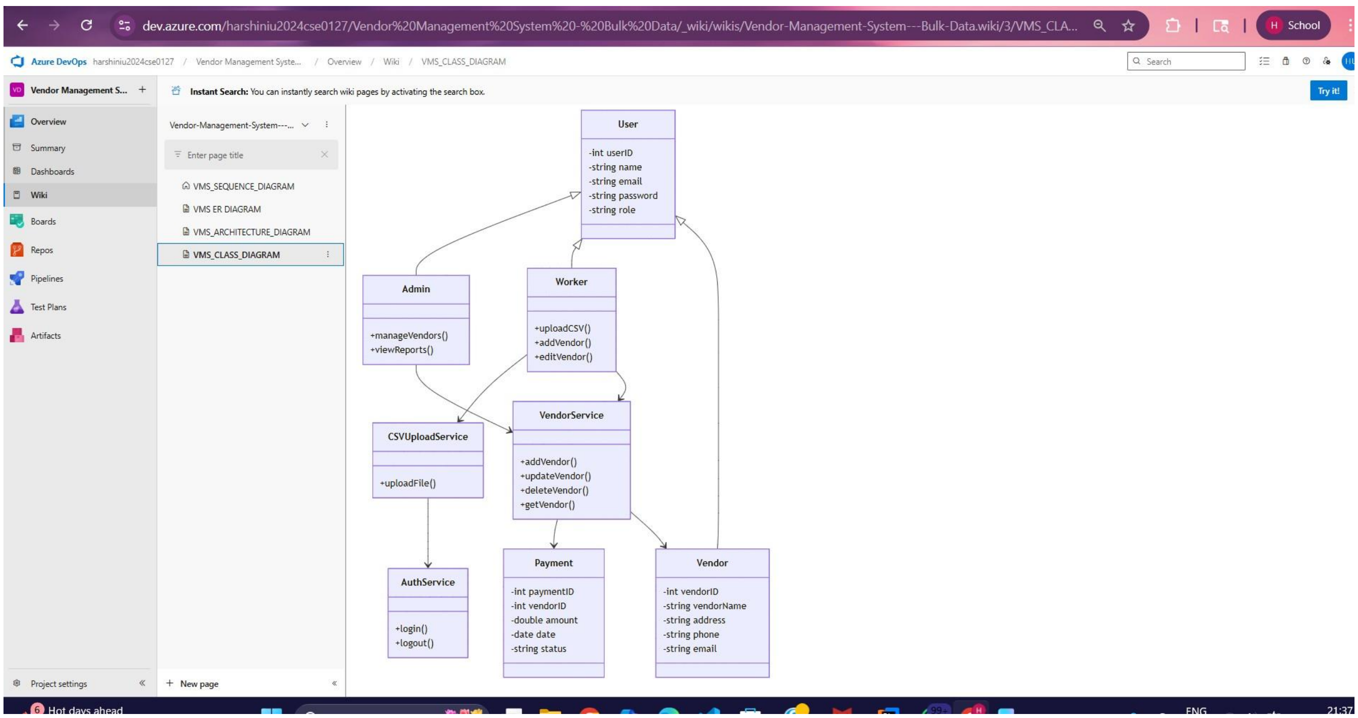
EXP NO:5	DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

Aim:

To Design a Class Diagram and Sequence Diagram for the given Project. 6A. Class

Diagram



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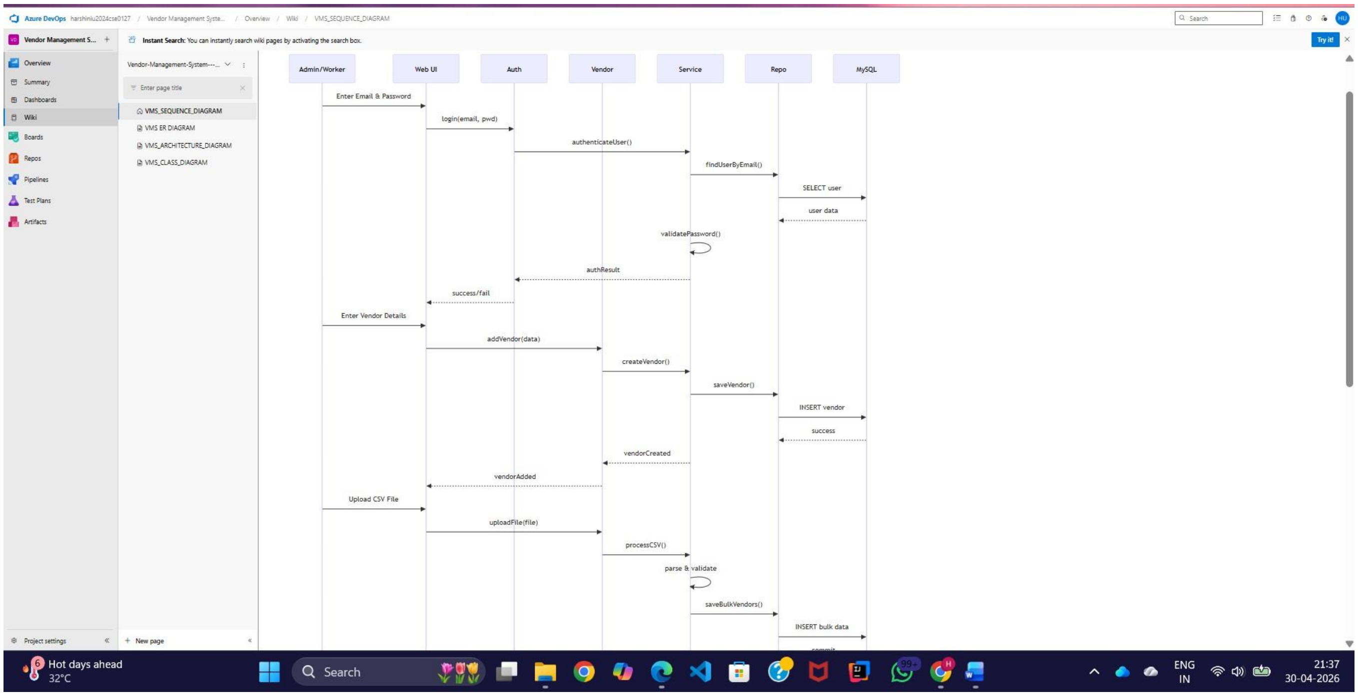
6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Product price Comparison System Project.

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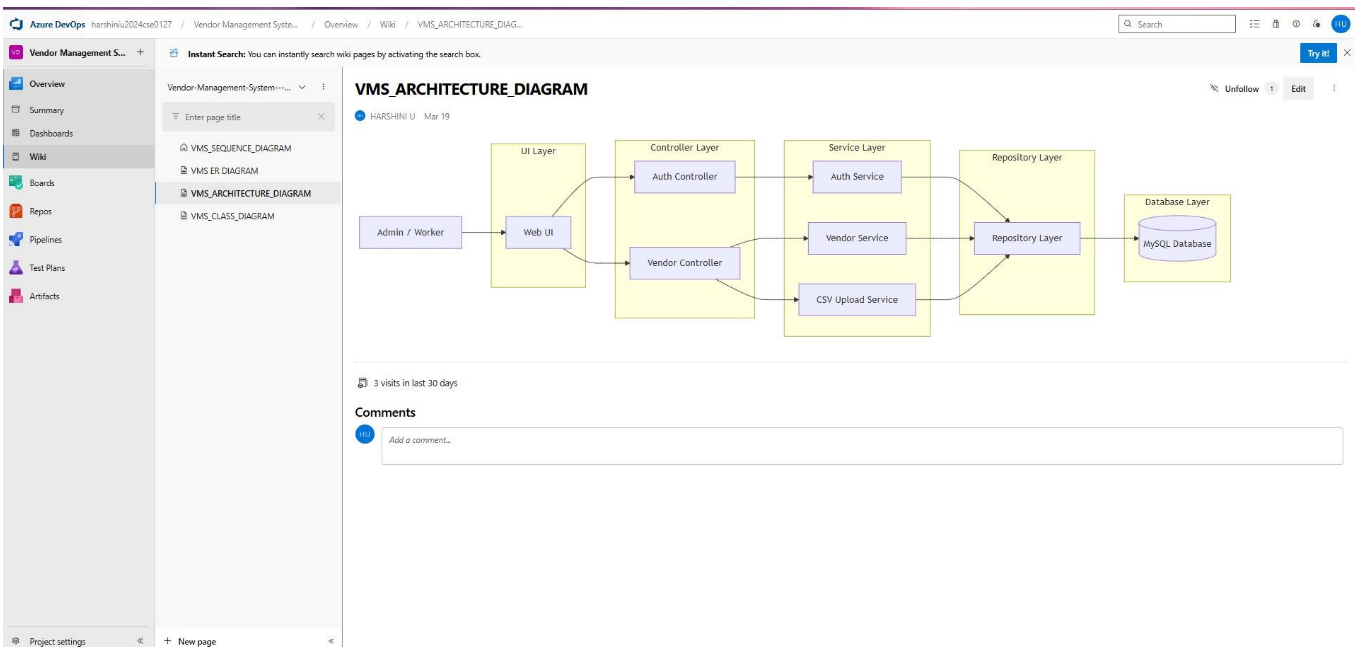
EXP NO: 7	DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

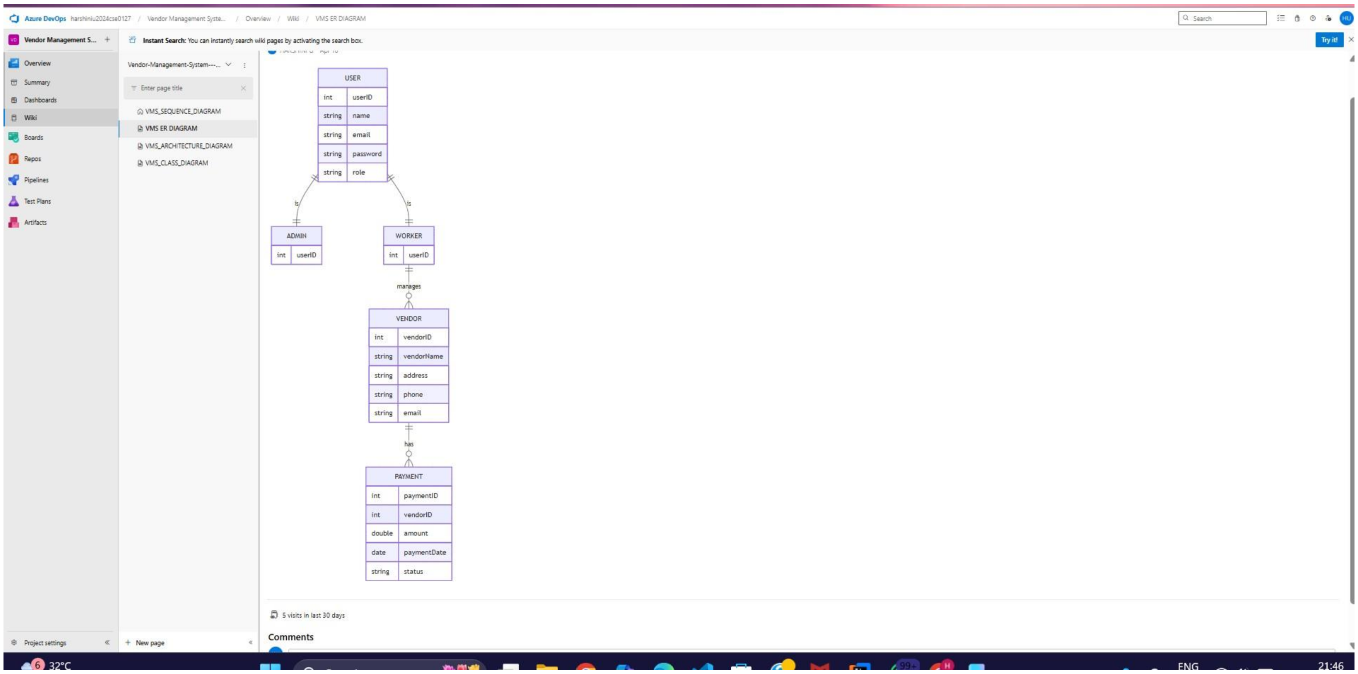
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project. 7A. Architectural

Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Product price Comparison System Project.

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EXP NO:8	TESTING – TEST PLANS AND TEST CASES

AIM

To design and implement test cases for the Vendor Management System using Azure DevOps, covering both **happy path (successful scenarios)** and **error scenarios** to ensure system reliability and correctness

TEST PLANNING AND TEST CASE DESIGN

1. Understand Core Features of the Application

User Authentication (Admin and Worker Login)

o Role-Based Access Control o Vendor

Management (Add/View Vendors) o

Bulk Vendor Data Upload (CSV) o

Dashboard Overview (Admin & Worker)

o Payment Tracking

2. Define User Interactions

Each test case simulates real user behavior such as:

o Logging into the system o Adding vendor details o

Uploading bulk vendor data

o Viewing dashboard information

3. Design Happy Path Test Cases

Focused on verifying that the system works correctly under normal conditions.

o **Examples:**

o User logs in with valid credentials o Worker successfully adds vendor details

o CSV file uploads and data is stored correctly

4. Design Error Path Test Cases

Covers invalid or unexpected scenarios to test system robustness.

o **Examples:**

o Login fails with invalid credentials o Vendor submission fails with

missing fields o CSV upload fails due to incorrect format

5. Break Down Steps and Expected Results Each test case includes:

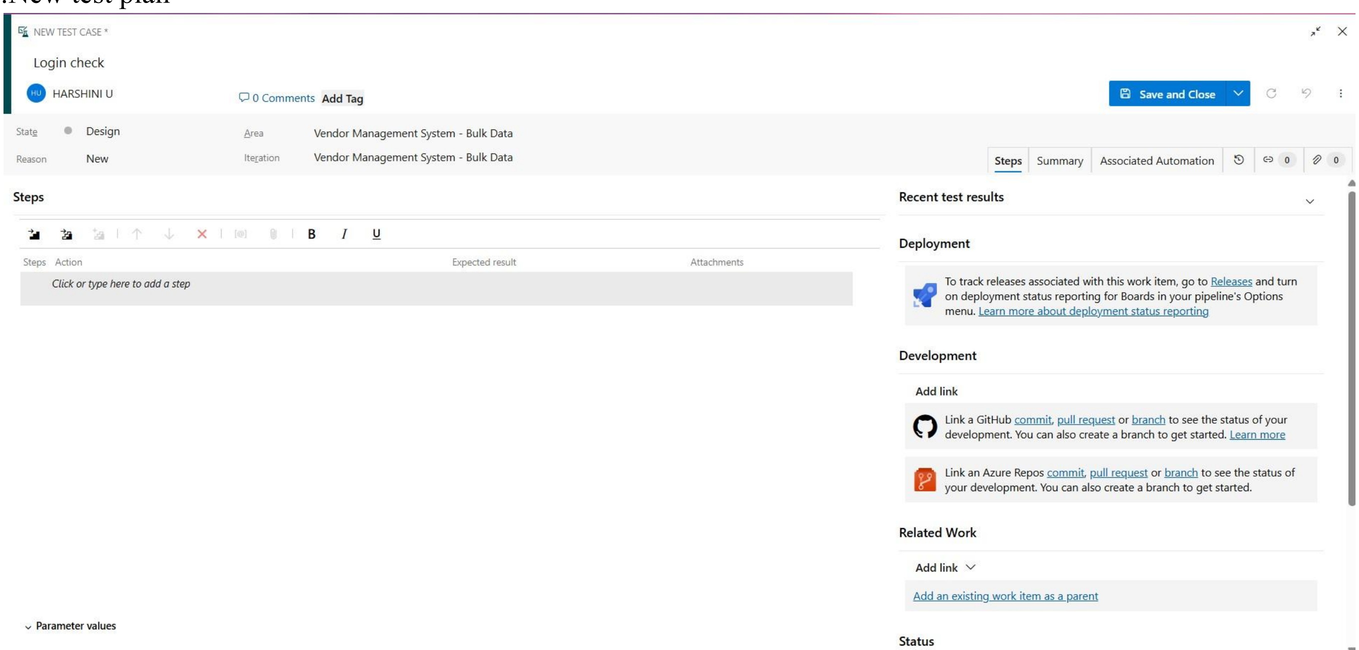
- o Step-by-step user actions
- o Corresponding expected outcomes
- o This ensures clarity and helps in proper validation of system behavior. o Reviewed for completeness and traceability against feature requirements.

6. Separate Test Suites

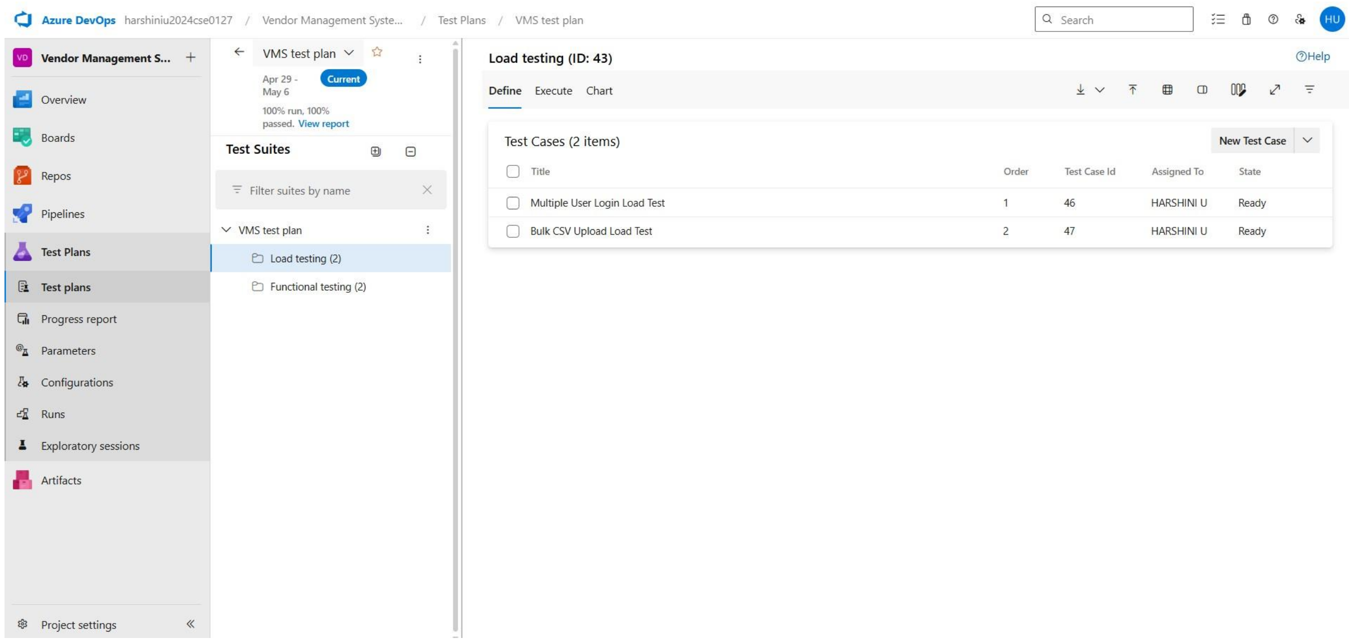
Test cases are grouped into suites for better organization:

- **Functional Testing** ○ Login Validation
  - ○ Add Vendor
- **Load Testing** ○ Multiple User Login
  - ○ Bulk Upload Handling

1.New test plan



2.Test suite



TEST CASES

Vendor Management System – Test Plans

USER STORIES

- **User Story 1:**
  As an admin/worker, I want to log into the system so that I can access the dashboard.
- **User Story 2:**
  As a worker, I want to add vendor details so that vendor data can be stored and managed.

TEST SUITES

Test Suite: TS01 – User Login

TC01 – Successful Login (Happy Path)

Action:

- Navigate to Login page
- Enter valid username and password
- Click "Login"

Expected Results:

- Login page loads correctly
- Input fields accept valid data
- User is authenticated successfully
- User is redirected to the dashboard

Type: Happy Path

TC02 – Login with Invalid Credentials (Error Scenario)

Action:

- Navigate to Login page
- Enter invalid username or password
- Click "Login"

Expected Results:

- Input fields accept data
- Error message like "Invalid credentials" is displayed
- User is not logged in
- Login page remains displayed

Type: Error Path

▧ Test Suite: TS02 – Vendor Management

TC03 – Add Vendor Successfully (Happy Path)

Action:

- Login as worker
- Navigate to Add Vendor page
- Enter valid vendor details (name, contact, etc.)
- Click "Submit"

Expected Results:

- Add Vendor page loads correctly
- All inputs are accepted
- Vendor details are stored in the database
- Success message is displayed
- Vendor appears in vendor list

Type: Happy Path

TC04 – Add Vendor with Missing/Invalid Data (Error Scenario)

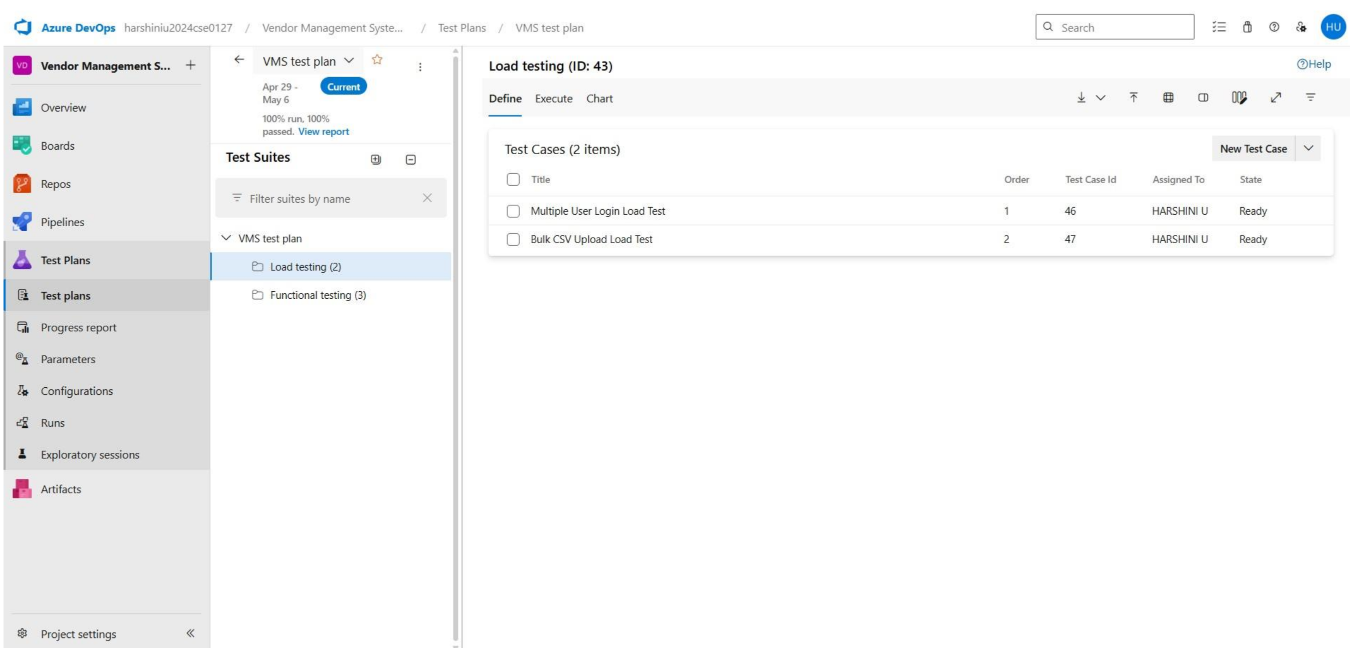
Action:

- Login as worker
- Navigate to Add Vendor page
- Leave required fields empty OR enter invalid data
- Click "Submit"

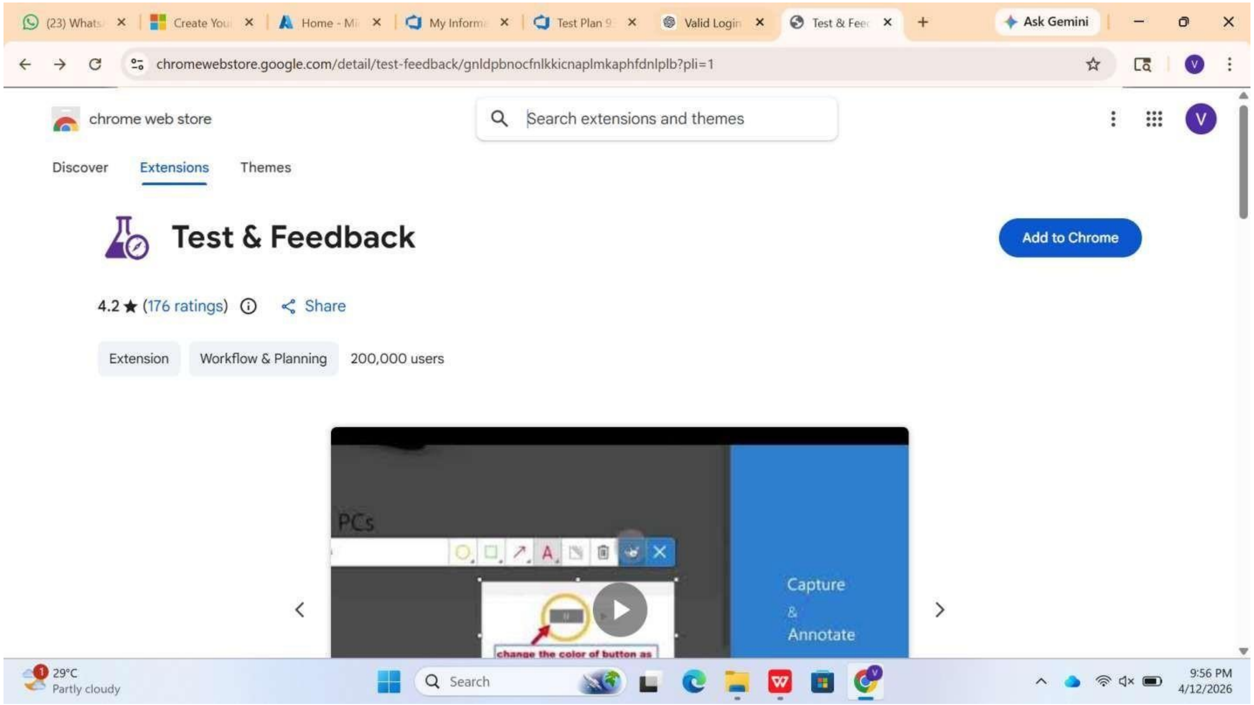
Expected Results:

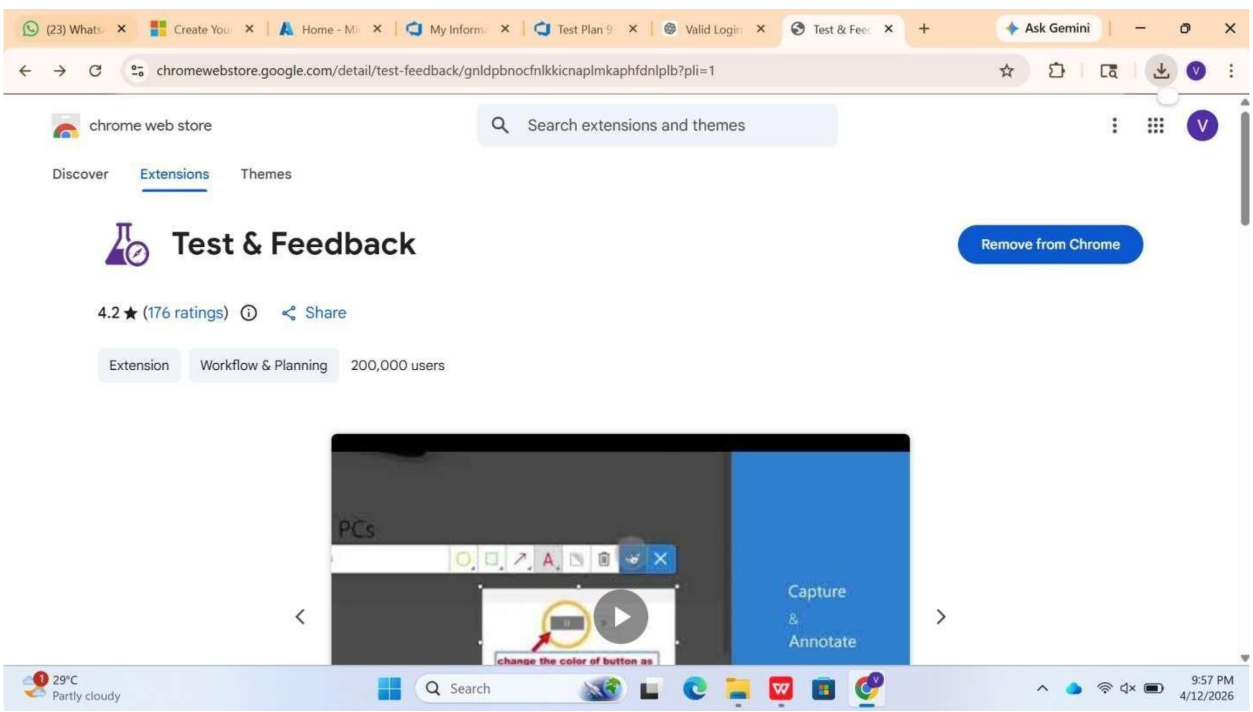
- System validates input fields
- Error messages are displayed (e.g., "Required field missing")
- Vendor is NOT added to database
- User remains on the same page
- No application crash occurs

Type: Error Path



4.Installation of test





Test and feedback
Showing it as an extension

5.Running the test cases

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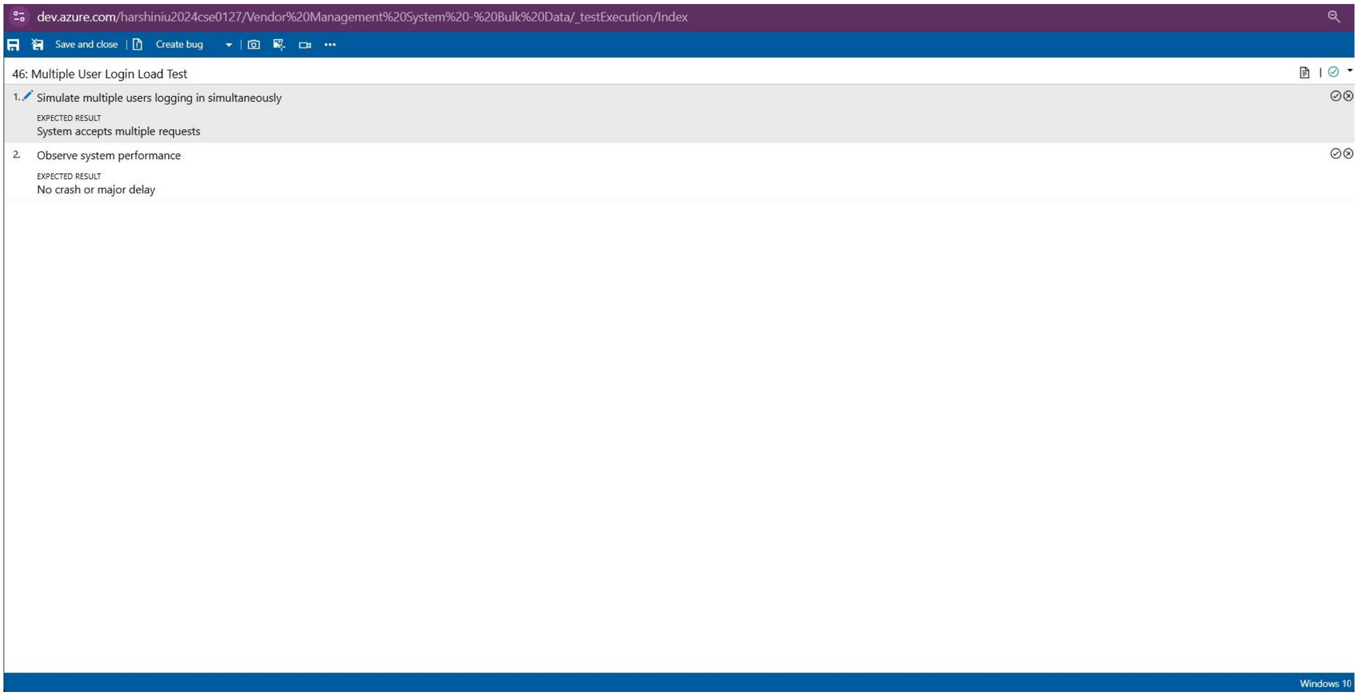
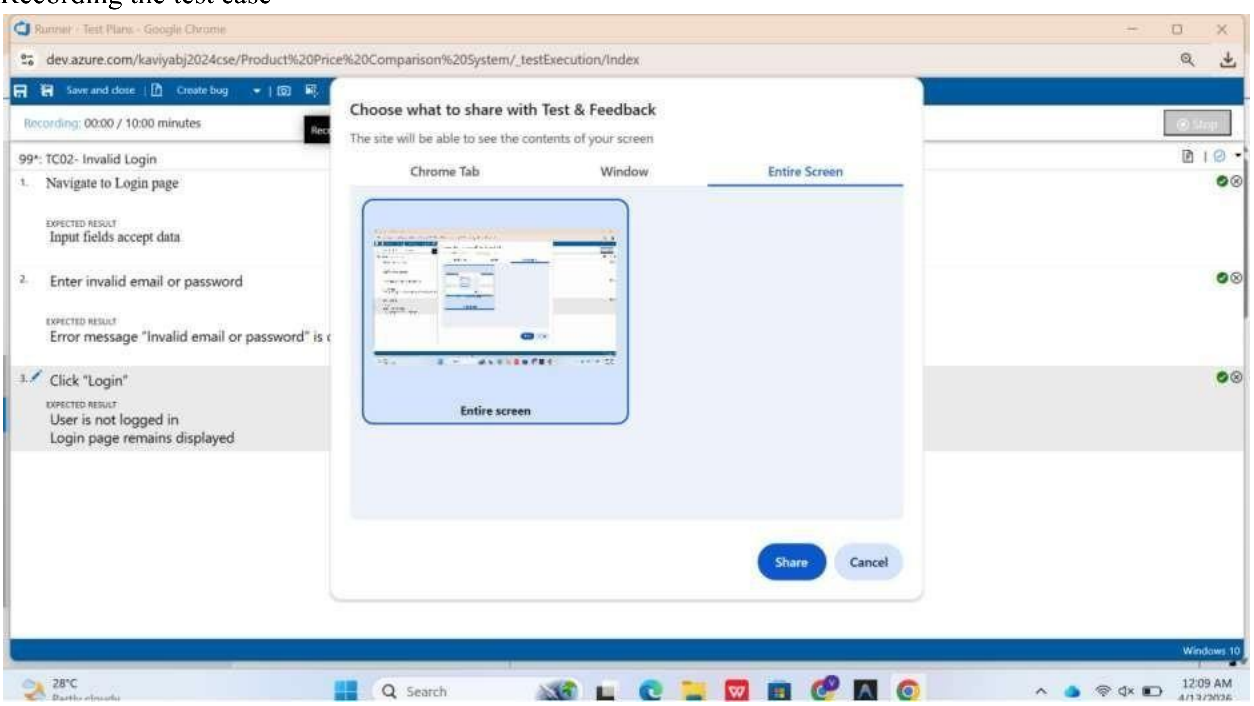


6.Recording the test case



7.Creating the bug

In our case no bugs yet

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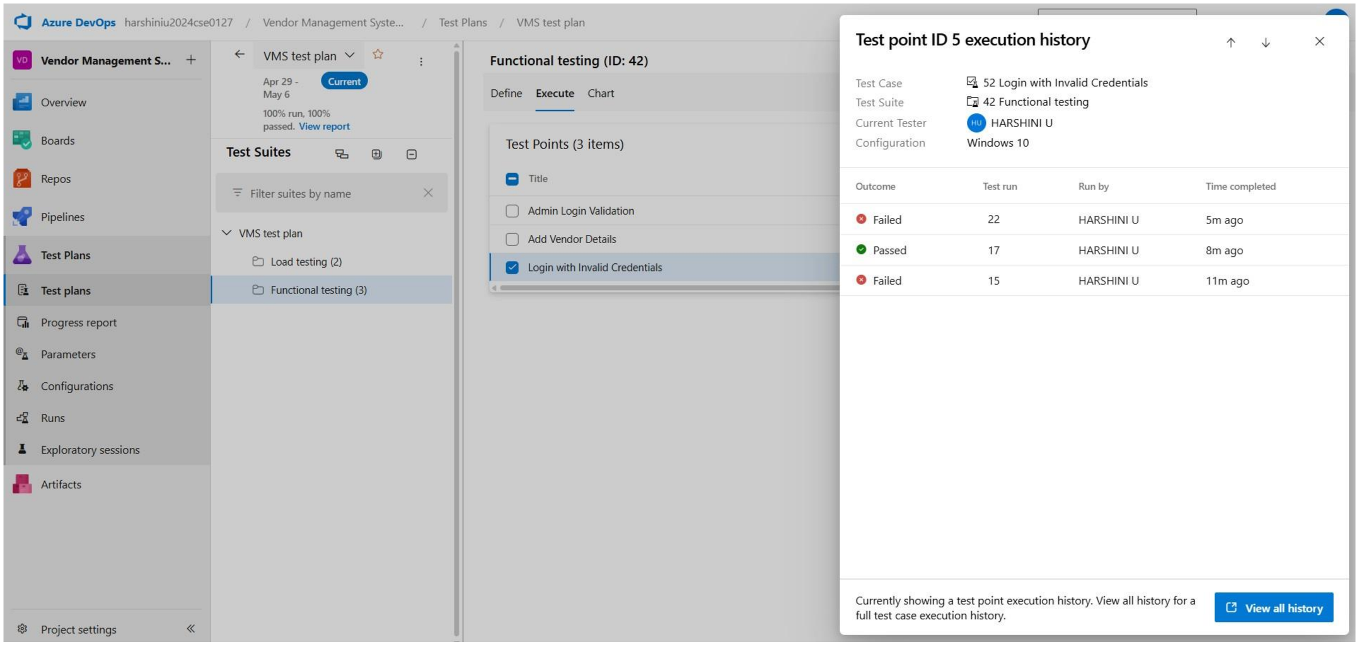
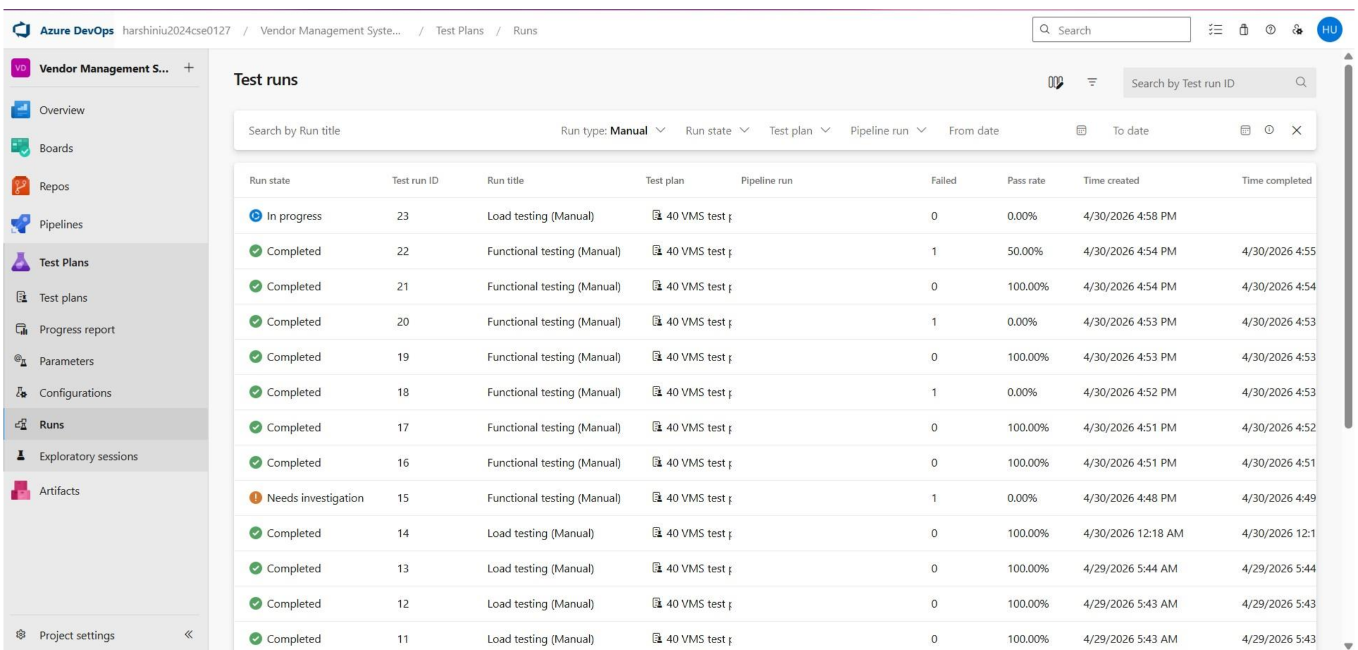
Just for example



8.Test case results



9. Test report summary

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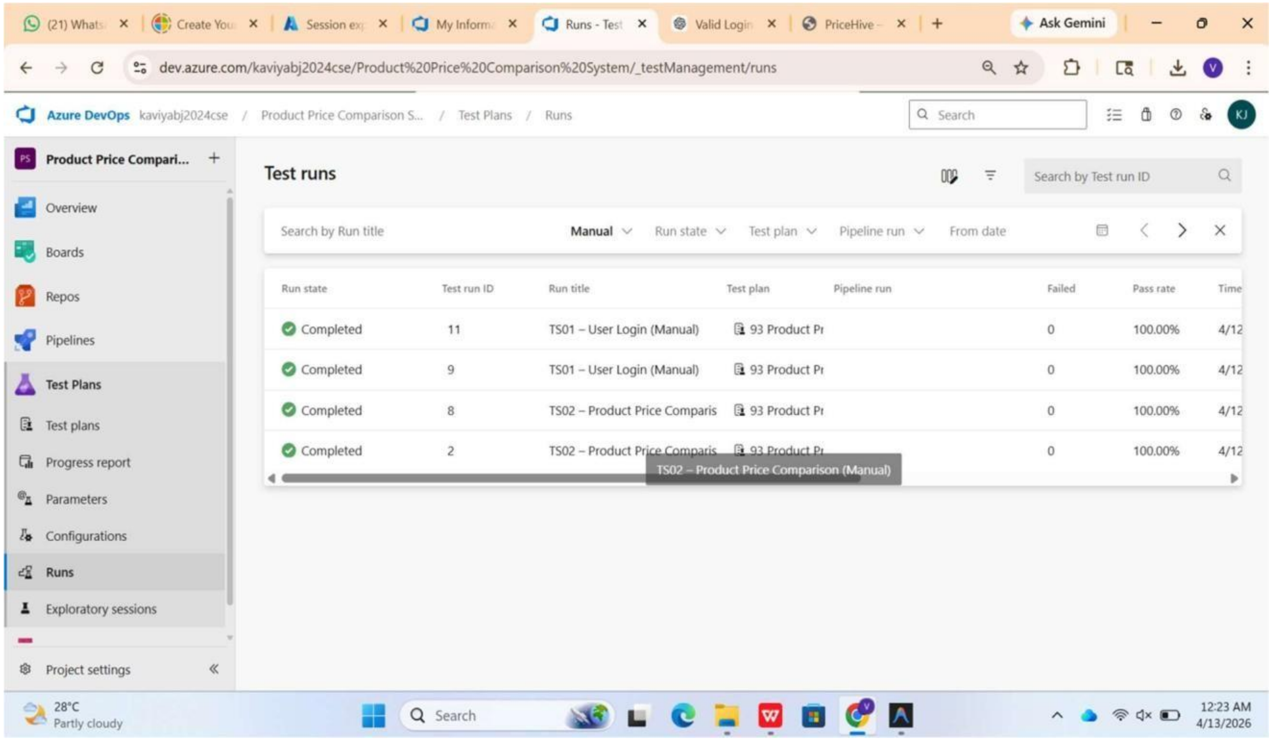
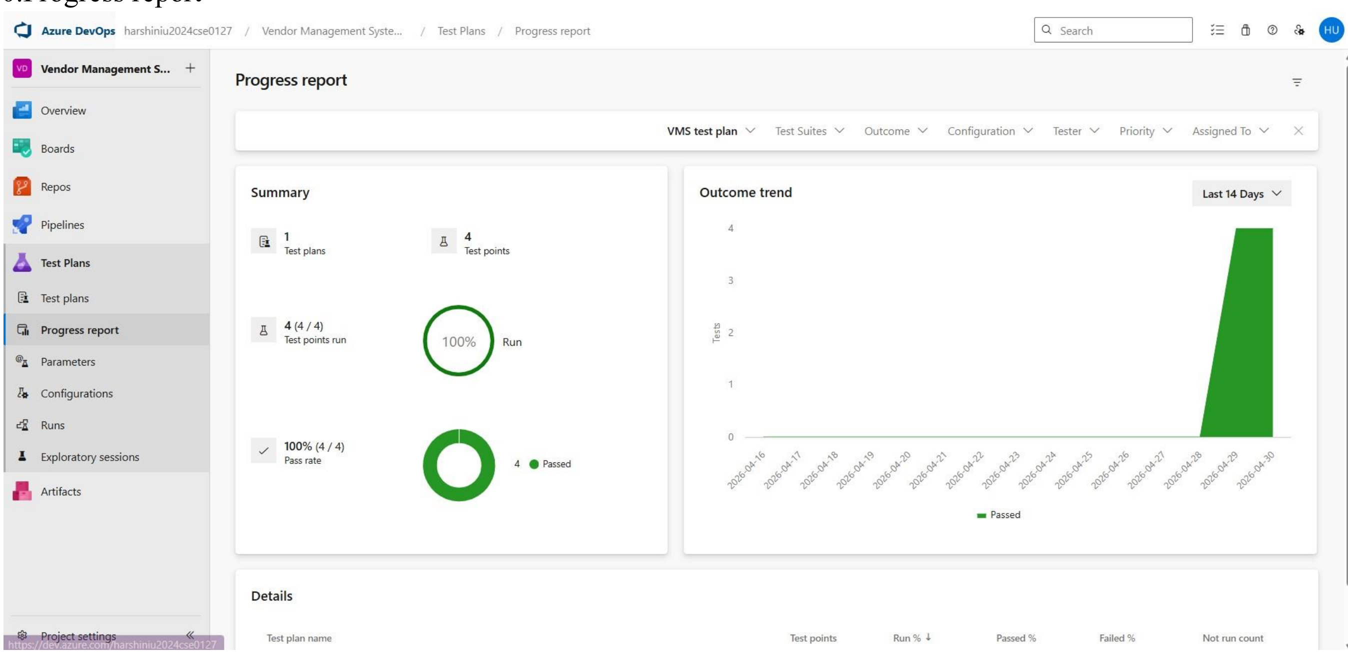


10.Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9	LOAD TESTING AND PIPELINES

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

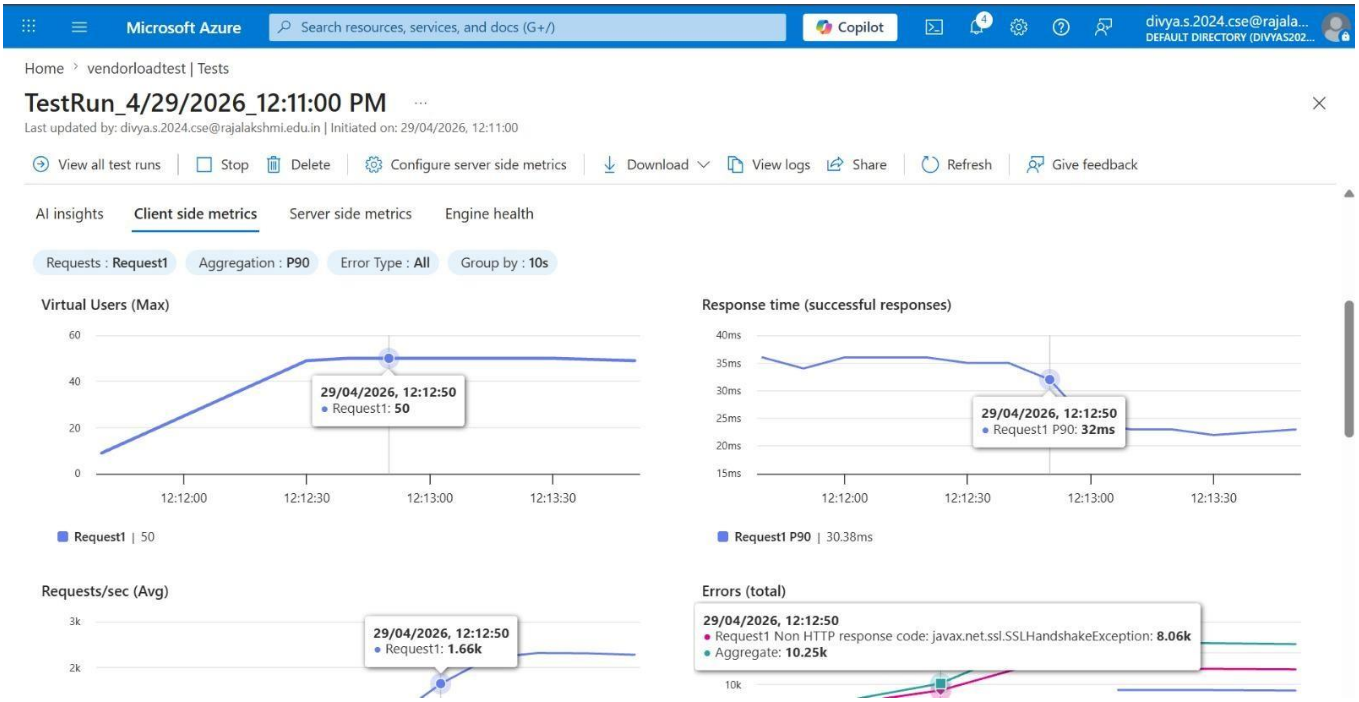
1. Sign in to Azure Portal
   Go to https://portal.azure.com and log in.
   Create a resource
2. Create the Resource o Go to → Search for "Azure Load Testing". o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details o Subscription: Choose your Azure subscription. o Resource Group: Create new or select an existing one. o Name: Provide a unique name (no special characters).
   - Location: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

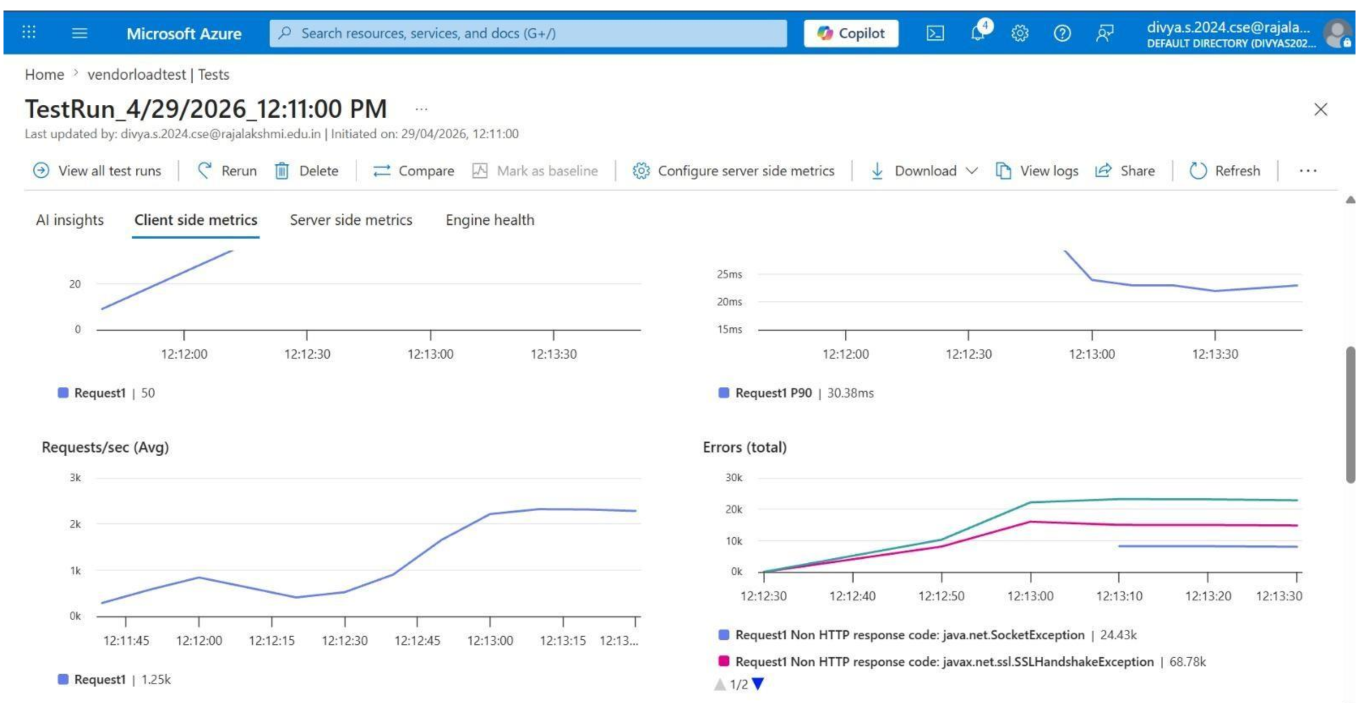
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab o Test Name: Provide a unique name.
   - Description: (Optional) Add test purpose.
   - Run After Creation: Keep checked.
3. Load Settings o Test URL: Enter the target endpoint (e.g., https://yourapi.com/products).
4. Click Review + Create → Create to start the test.

Load Testing





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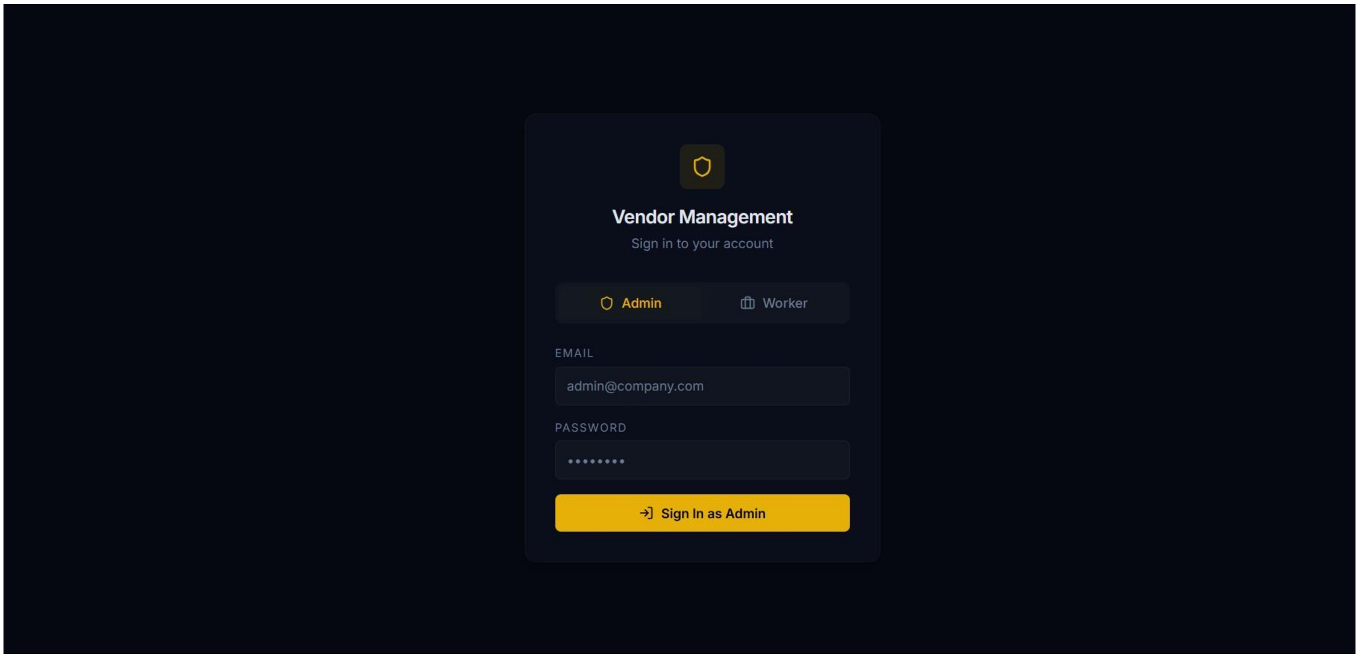

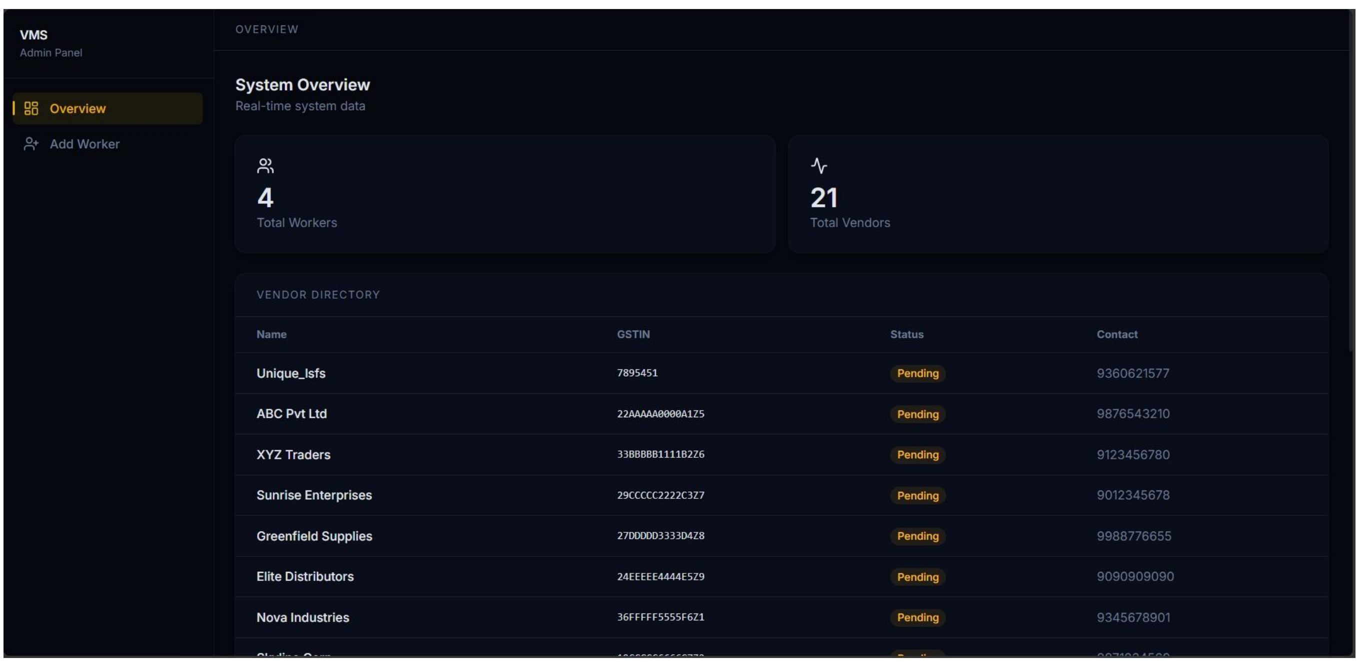

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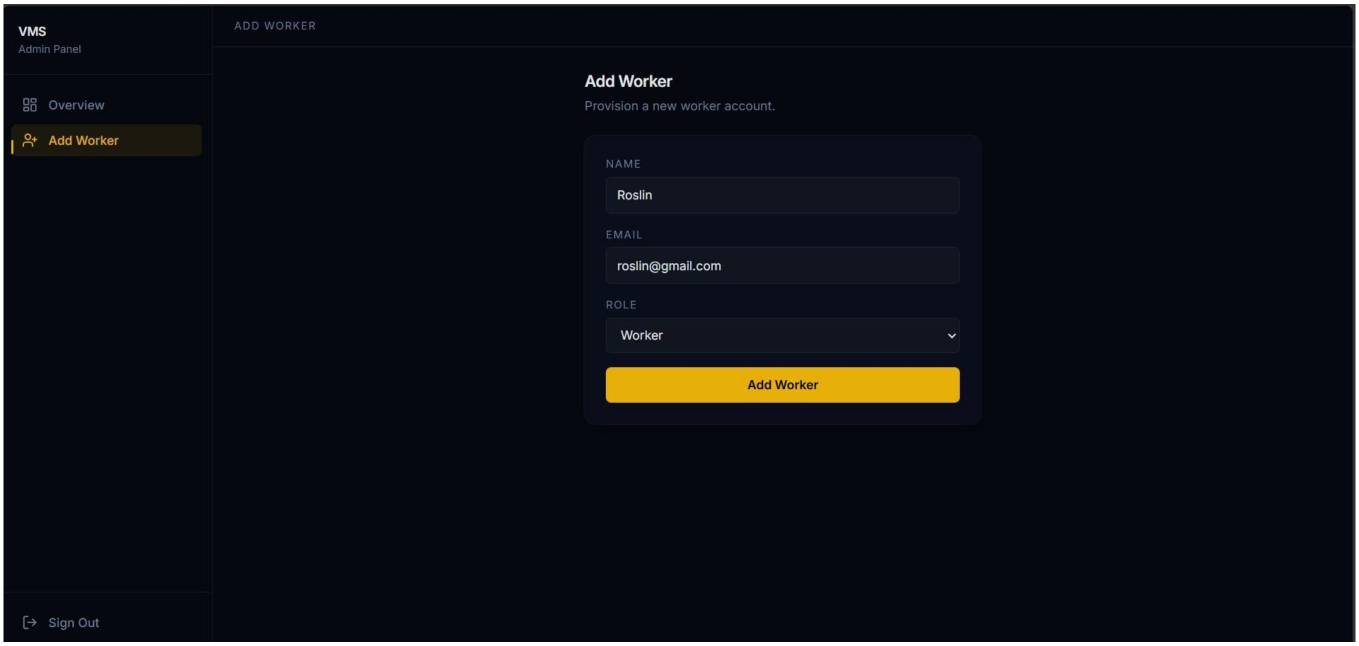

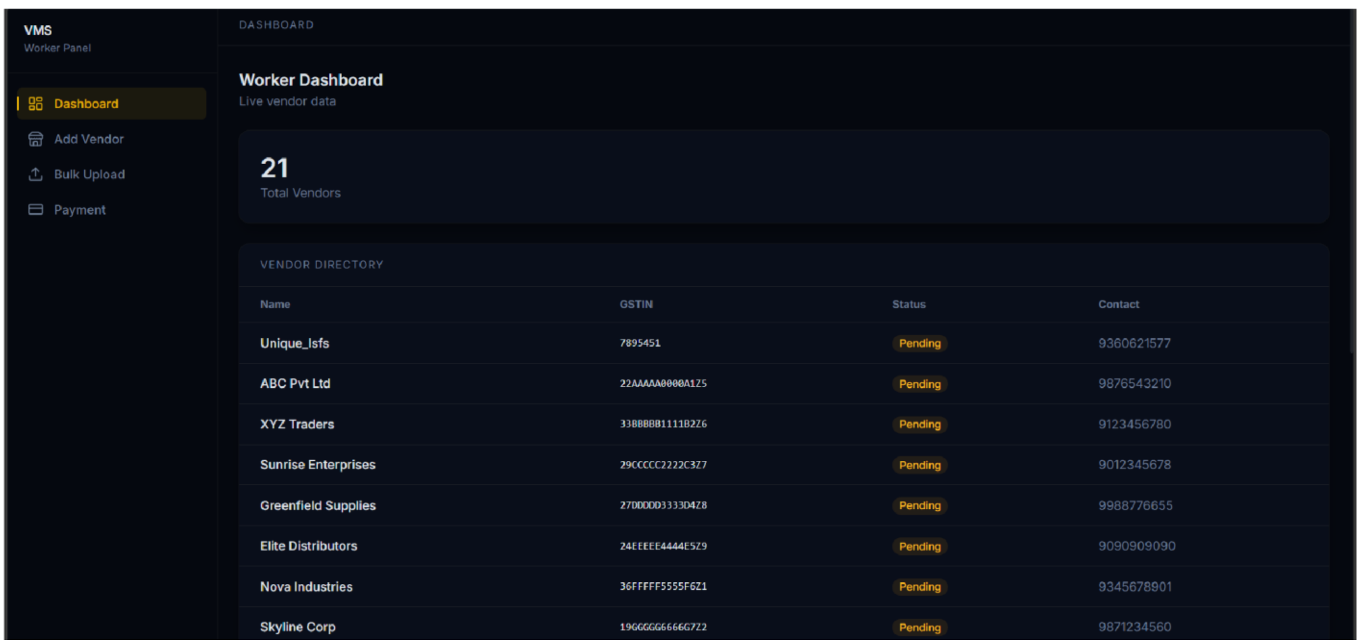

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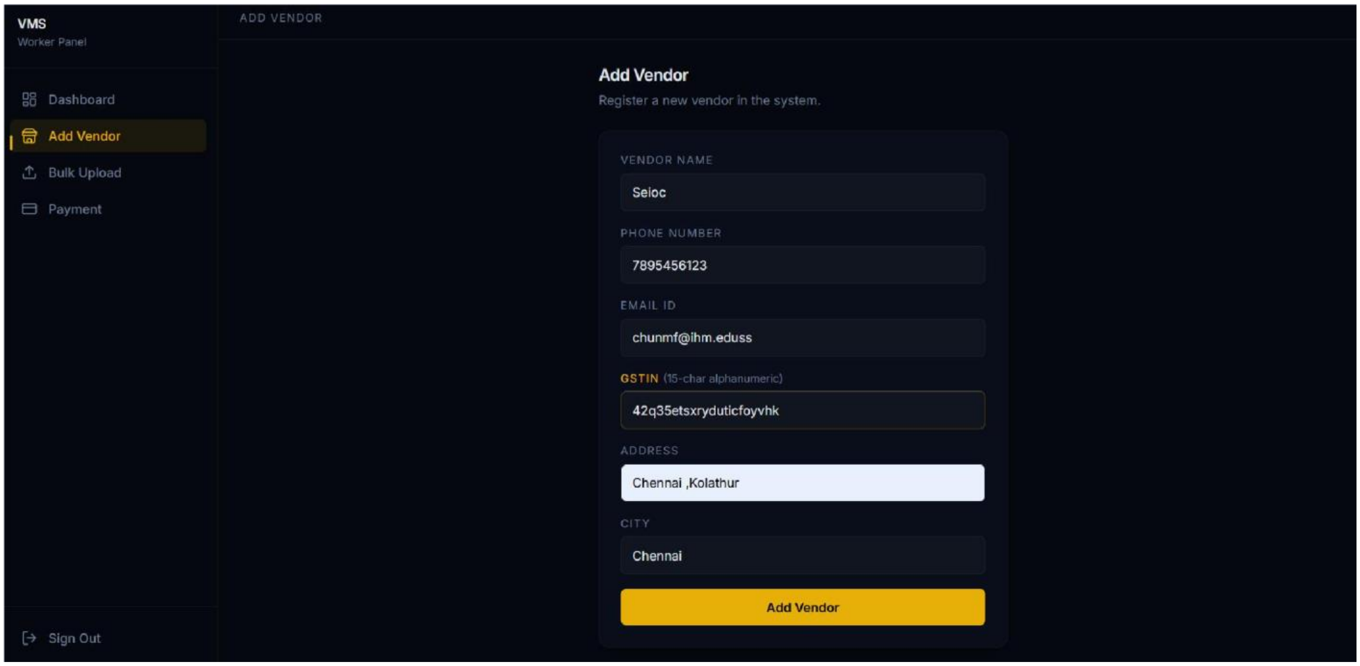

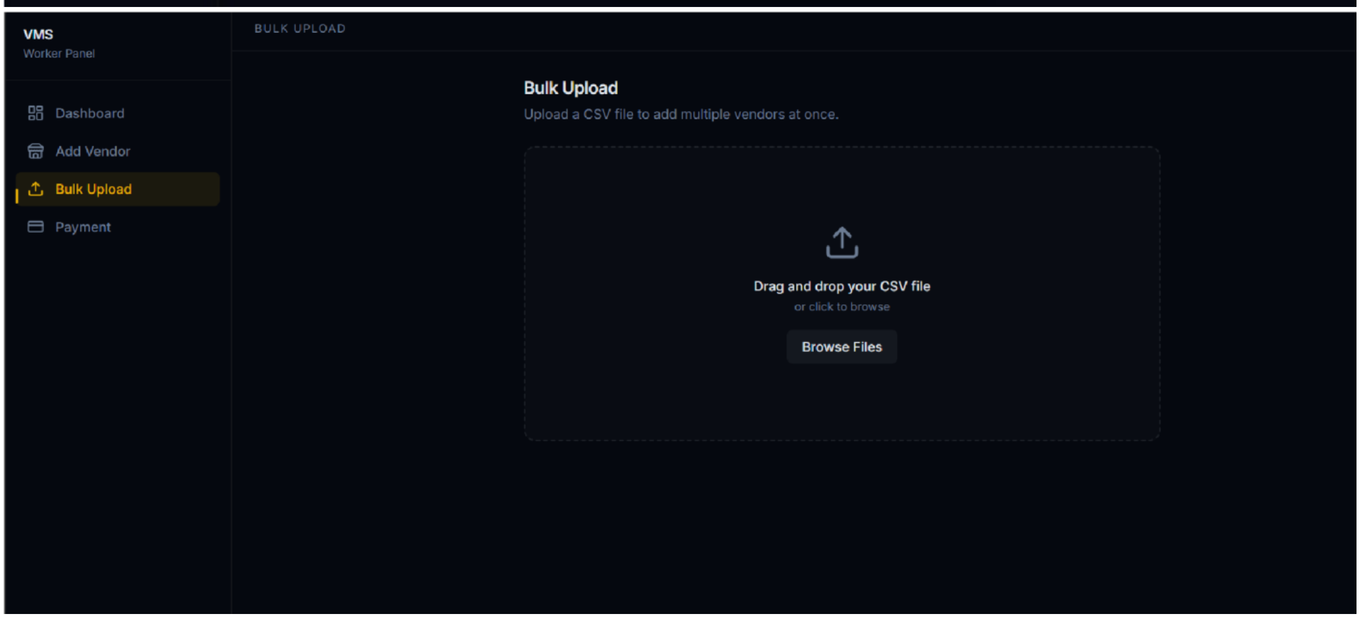

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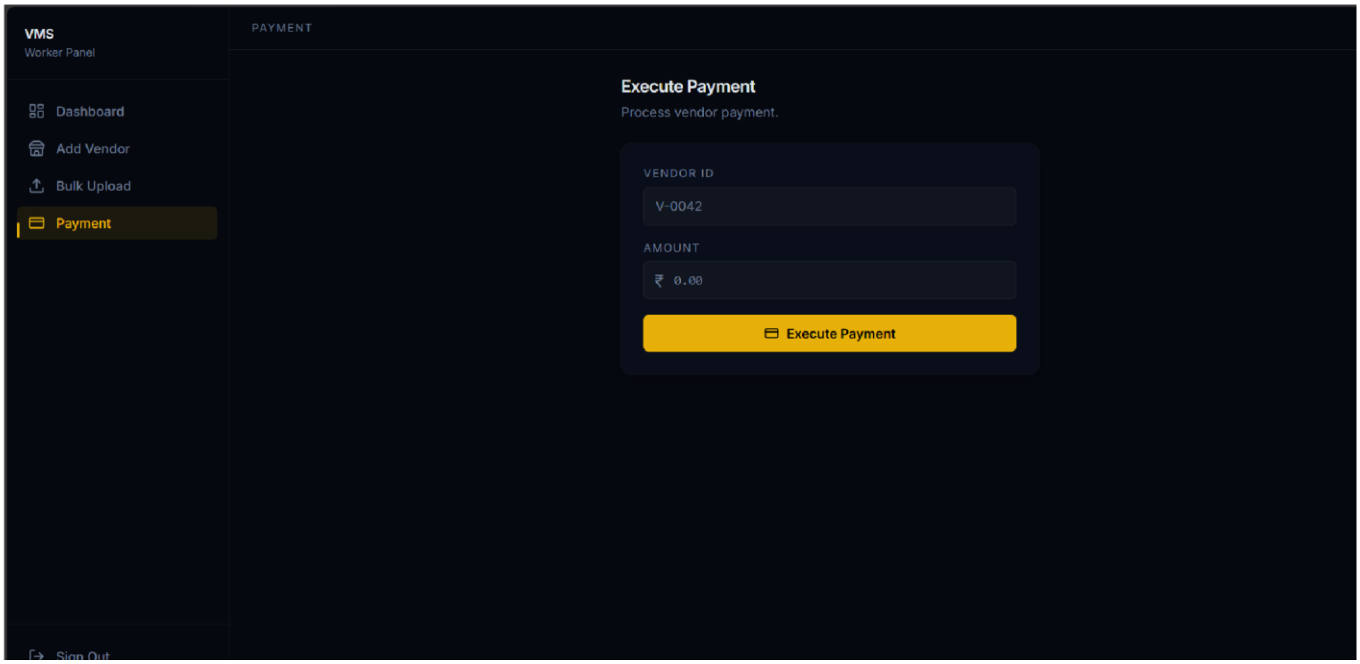
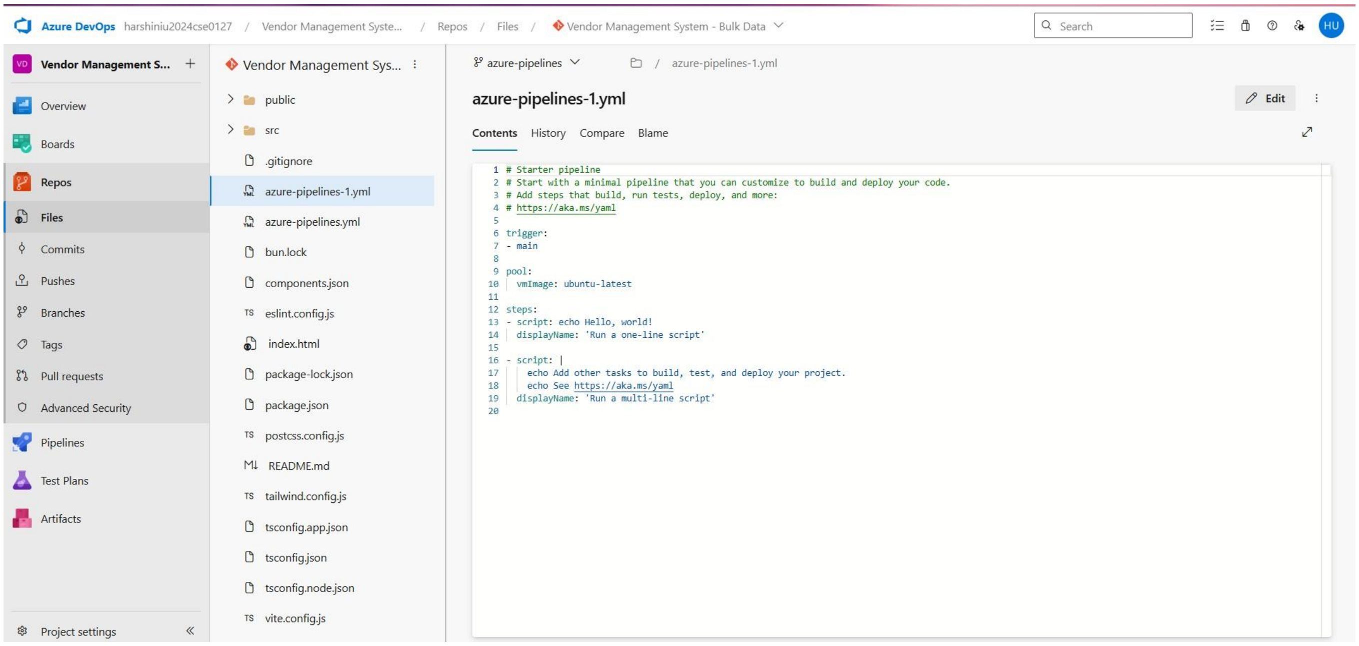


Pipelines

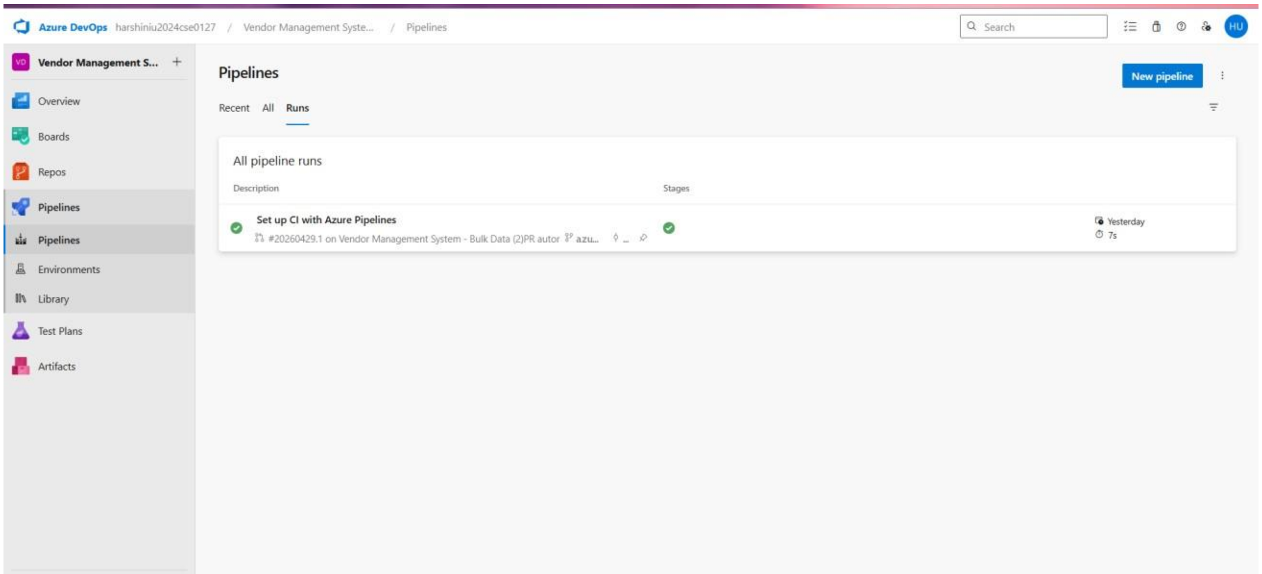
Steps:

1. Connect GitHub to Azure DevOps: o In Azure DevOps, create a new project.
    - o Create a pipeline and select GitHub as the source.
    - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root: o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
    - o Add the following basic pipeline configuration

yml Code


Azure DevOps
harshiniu2024cse0127
Vendor Management Syste...
Repos
Files
Vendor Management System - Bulk Data
Search
Vendor Management S...
Overview
Boards
Repos
Files
Commits
Pushes
Branches
Tags
Pull requests
Advanced Security
Pipelines
Test Plans
Artifacts
Project settings
Vendor Management Sys...
public
src
.gitignore
azure-pipelines-1.yml
azure-pipelines.yml
bun.lock
components.json
eslint.config.js
index.html
package-lock.json
package.json
postcss.config.js
README.md
tailwind.config.js
tsconfig.app.json
tsconfig.json
tsconfig.node.json
vite.config.js
azure-pipelines
azure-pipelines-1.yml
azure-pipelines-1.yml
Edit
Contents
History
Compare
Blame
Starter pipeline
Start with a minimal pipeline that you can customize to build and deploy your code.
Add steps that build, run tests, deploy, and more:
https://aka.ms/yaml
trigger:
- main
pool:
vmImage: ubuntu-latest
steps:
- script: echo Hello, world!
displayName: 'Run a one-line script'
- script: |
echo Add other tasks to build, test, and deploy your project.
echo See https://aka.ms/yaml
displayName: 'Run a multi-line script'

Pipeline



Result:

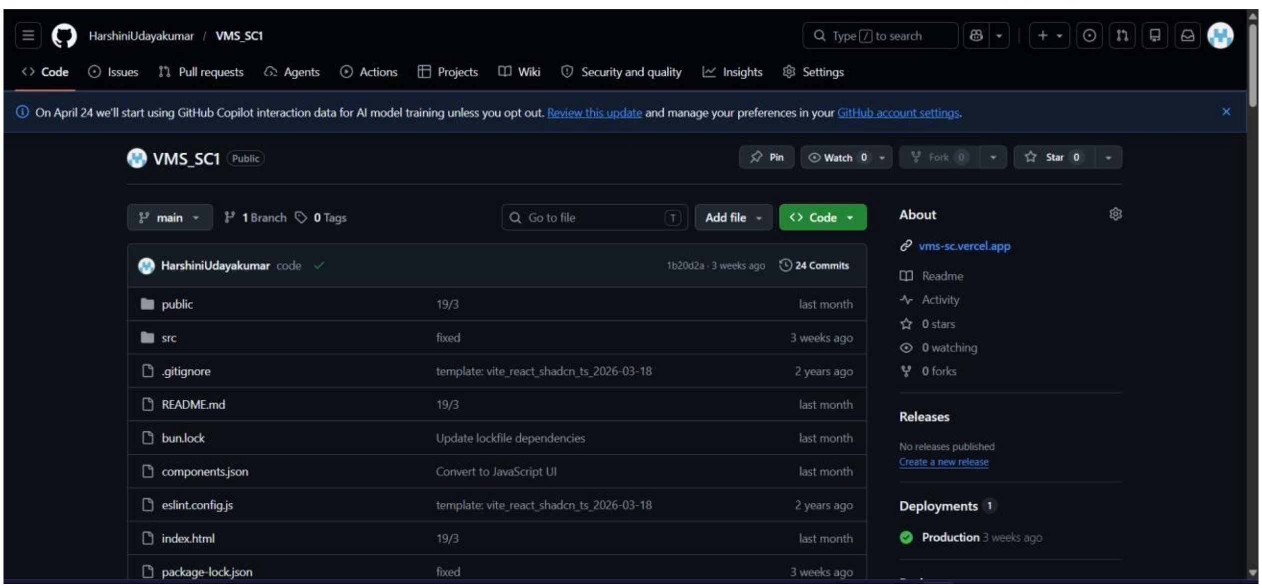
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

EXP NO:10	GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Product Price Comparison System project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.